Classical Mechanics
Post - Midsem Notes
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If we start with Lagrangian of the form:

$$\mathcal{L} = \mathcal{L}_0 + \underbrace{a_i \dot{q}_i}_{} + \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$$

→ Not easy to get this term:
▷ Non-trivial, time-dependent Transform

▫ Note: Showing non-trivial Time-Dependence [Not done in class, extra]

We start from Kinetic Energy in some coordinates Q_i:

$$T = \frac{1}{2} T_{ij} (Q,t) \dot{Q}_i \dot{Q}_j$$

Now, a time-dependent change of generalized coordinates:

$$Q_i = Q_i(q,t)$$

$$\Rightarrow \dot{Q}_i = \frac{\partial Q_i}{\partial q_k} \dot{q}_k + \frac{\partial Q_i}{\partial t}$$

▫ Substituting into T:

$$T = \frac{1}{2} T_{ij} (Q,t) \left(\frac{\partial Q_i}{\partial q_k} \dot{q}_k + \frac{\partial Q_i}{\partial t} \right) \left(\frac{\partial Q_j}{\partial q_l} \dot{q}_l + \frac{\partial Q_j}{\partial t} \right)$$

$$= \frac{1}{2} \left(T_{ij} \frac{\partial Q_i}{\partial q_k} \frac{\partial Q_j}{\partial q_l} \right) \dot{q}_k \dot{q}_l + \left(T_{ij} \frac{\partial Q_i}{\partial q_k} \frac{\partial Q_j}{\partial t} \right) \dot{q}_k + \frac{1}{2} \left(T_{ij} \frac{\partial Q_i}{\partial t} \frac{\partial Q_j}{\partial t} \right)$$

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We identify: $\boxed{\tilde{T}_{kl}(q,t) = T_{ij}(Q,t)\frac{\partial Q_i}{\partial q_k}\frac{\partial Q_j}{\partial q_l}}$

$$\boxed{a_k(q,t) = T_{ij}(Q,t)\frac{\partial Q_i}{\partial q_k}\frac{\partial Q_j}{\partial t}}$$

The last piece $\frac{1}{2}T_{ij}(\partial_t Q_i)(\partial_t Q_j)$ behaves like a potential (no velocities $\dot{q}$).

The term $a_k(q,t)\dot{q}_k$ term appears only when:

$$\boxed{\frac{\partial Q_i}{\partial t} \neq 0} \rightarrow \text{The coordinate transform is non-trivially time-dependent.}$$

——— ◦ End of Extra Notes ◦ ———

$$L_0 = L_0(q,t) + a_i(q,t)\dot{q}_i + \frac{1}{2}T_{ij}(q,t)\dot{q}_i\dot{q}_j$$

With $T_{ij} = T_{ji}$.

The canonical momentum is:

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = a_i + T_{ij}\dot{q}_j$$

Solving this for the velocities:

$$\boxed{T_{ij}\dot{q}_j = p_i - a_i} \Rightarrow \boxed{\dot{q}_j = T^{-1}_{jk}(p_k - a_k)}$$

▫ Extra ▹ How did this transform come about?

[P.T.O.]

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Let T^{-1}_{jk} be inverse of T_{ij}, satisfying

$$T_{ij} T^{-1}_{jk} = \delta_{ik}$$

So, $T^{-1}_{kj} T_{ji} \dot{q}_i = T^{-1}_{kj} (P_j - a_j)$

$$\Rightarrow \delta_{ki} \dot{q}_i = T^{-1}_{kj} (P_j - a_j)$$

Renaming the dummy index $i \to k$ on the ~~fo~~ left:

$$\dot{q}_k = T^{-1}_{kj} (p_j - a_j)$$

All repeated indices are dummy indices:

$$\dot{q}_j = T^{-1}_{jk} (P_k - a_k), \quad \dot{q}_i = T^{-1}_{ij} (P_j - a_j)$$

$$\therefore \boxed{T_{ij} \dot{q}_j = P_i - a_i \Rightarrow \dot{q}_j = T^{-1}_{jk} (P_k - a_k)}$$

- Back to Class Notes:

Hamiltonian is : $H = P_i \dot{q}_i - L_c$

Substituting L_c and grouping terms:

$$H = P_i \dot{q}_i - \left(L_0 + a_k \dot{q}_k + \frac{1}{2} T_{mn} \dot{q}_m \dot{q}_n\right)$$

$$= (P_i - a_i) \dot{q}_i - \frac{1}{2} T_{mn} \dot{q}_m \dot{q}_n - L_0$$

Using $p_i - a_i = T_{ij} \dot{q}_j$. Note the identity:

$(P_i - a_i) \dot{q}_i = T_{ij} \dot{q}_j \dot{q}_i = 2 \cdot \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$ [P.T.O]

Thus, $(p_i - a_i)\dot{q}_i - \frac{1}{2}T_{ij}\dot{q}_i\dot{q}_j = \frac{1}{2}T_{ij}\dot{q}_i\dot{q}_j$

We substitute: $\boxed{\dot{q}_j = T^{-1}_{jk}(p_k - a_k)}$

directly into the combination to get the Hamiltonian purely in p, q:

$$H = (p_i - a_i)\dot{q}_i - \frac{1}{2}T_{ij}\dot{q}_i\dot{q}_j - \mathcal{L}_0$$

$$= \frac{1}{2}(p_i - a_i)\dot{q}_i - L_0$$

$$= \frac{1}{2}(p_i - a_i)T^{-1}_{ij}(p_j - a_j) - \mathcal{L}_0$$

$$\boxed{H = \frac{1}{2}T^{-1}_{ij}(p_i - a_i)(p_j - a_j) - \mathcal{L}_0}$$

Extra Note: Explanation of substitution
(not in class)

$$\dot{q}_j = T^{-1}_{jk}(p_k - a_k)$$

We start from $\mathcal{L} = \mathcal{L}_0 + a_i\dot{q}_i + \frac{1}{2}T_{ij}\dot{q}_i\dot{q}_j$

and the canonical momentum:

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = a_i + T_{ij}\dot{q}_j$$

So, $T_{ij}\dot{q}_j = p_i - a_i$

Now, such that, $T^{-1}_{ik}T_{kj} = \delta_{ij}$, we solve for the velocities:

$$\dot{q}_l = T^{-1}_{lm}(p_m - a_m)$$

Now, $H = p_i \dot{q}_i - L_0$

Substitute L_0,

$$H = p_i \dot{q}_i - \left(L_{00} + a_j \dot{q}_j + \frac{1}{2} T_{kl} \dot{q}_k \dot{q}_l\right)$$

$$\Rightarrow H = (p_i - a_i)\dot{q}_i - \frac{1}{2} T_{kl} \dot{q}_k \dot{q}_l - L_{00}$$

Substituting, $\dot{q}_i = T^{-1}_{ir}(p_r - a_r)$ everywhere:

▷ 1st term:

$$(p_i - a_i)\dot{q}_i = (p_i - a_i) T^{-1}_{ir} (p_r - a_r)$$

▷ 2nd term:

$$\frac{1}{2} T_{kl} \dot{q}_k \dot{q}_l = \frac{1}{2} T_{kl} \left(T^{-1}_{ks}(p_s - a_s)\right)\left(T^{-1}_{lt}(p_t - a_t)\right)$$

Using $T_{kl} T^{-1}_{ks} = \delta_{ls}$ (or equivalently $T_{kl} T^{-1}_{lt} = \delta_{kt}$)

$$T_{kl} T^{-1}_{ks} T^{-1}_{lt} = T^{-1}_{st}$$

Because, $T_{kl} T^{-1}_{lt} = \delta_{kt}$ and Then $\delta_{kt} T^{-1}_{ts} = T^{-1}_{ts}$, thus:

$$\frac{1}{2} T_{kl} \dot{q}_k \dot{q}_l = \frac{1}{2} T^{-1}_{st} (p_s - a_s)(p_t - a_t)$$

So the Hamiltonian becomes:

$$H = (p_i - a_i) T^{-1}_{ir} (p_r - a_r) - \frac{1}{2} T^{-1}_{st} (p_s - a_s)(p_t - a_t) - L_{00}$$

$$\Rightarrow \boxed{H = \frac{1}{2} T^{-1}_{ij} (p_i - a_i)(p_j - a_j) - L_{00}}$$

—— End of Extra Note ——

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- <u>Back to Classnotes:</u>

▷ a, T_{ij} and $\mathcal{L}_0$ $\equiv$ are functions of q, t.

▷ H is function of p, q, t.

Let us consider a Lagrangian:

$$\boxed{\mathcal{L} = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right) + m\Omega r^2\sin^2\theta\,\dot{\phi} + \frac{1}{2}m\Omega^2 r^2\sin^2\theta - V(r)}$$

Here, $\boxed{\frac{1}{2}T_{ij}\dot{q}_i\dot{q}_j = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right)}$

$$\boxed{a_i\dot{q}_i = m\Omega r^2\sin^2\theta\,\dot{\phi}}$$

$$\boxed{\mathcal{L}_0 = \frac{1}{2}m\Omega^2 r^2\sin^2\theta - V(r)}$$

this is

Now, ~~let us consider~~ a particle in rotating frame with angular velocity Ω (constant), rotating about $\hat{z}$ axis.

▷ We have the <u>inertial frame Lagrangian</u>:

$$\mathcal{L}_{in} = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right) - V(r)$$

$\dot{\phi}$ = Azimuthal angular velocity in inertial frame.

□ We define <u>Rotating Frame</u>: [r, θ unchanged]

$$\phi' = \phi + \Omega t \Rightarrow \dot{\phi}' = \dot{\phi} + \Omega$$

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So, $L_{rotational} = \frac{m}{2}\left[\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,(\dot{\phi}' + \Omega)^2\right] - V(r)$

$$\Rightarrow \boxed{L_{rotational} = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}'^2\right) + m\Omega r^2\sin^2\theta\,\dot{\phi}' + \frac{m}{2}\Omega^2 r^2\sin^2\theta - V(r)}$$

Which is the expression we initially considered.

▫ Note: This was done because in order to study Newton's laws of motion in non-inertial frames, we need to have an inertial frame and a non-inertial frame with respect to it.

Now, $T_{ij} = m\begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix}$

$\Rightarrow T_{rr} = m, \quad T_{\theta\theta} = mr^2, \quad T_{\phi\phi} = mr^2\sin^2\theta$

[Standard diagonal kinetic energy matrix] → [Extra note.]

▸ The rows and columns correspond to coordinates : $q = (r, \theta, \phi)$

$T_{ij} = m\begin{pmatrix} T_{rr} & T_{r\theta} & T_{r\phi} \\ T_{\theta r} & T_{\theta\theta} & T_{\theta\phi} \\ T_{\phi r} & T_{\phi\theta} & T_{\phi\phi} \end{pmatrix}$ [Extra Note]

Now Kinetic Energy in general:

$$T = \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j$$

Plugging in T_{ij}

$$T = \frac{m}{2} \begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2 \sin^2\theta \end{pmatrix} \begin{pmatrix} \dot{r}^2 \\ \dot{\theta}^2 \\ \dot{\phi}^2 \end{pmatrix}$$

$$\boxed{T = \frac{m}{2}\left[\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}^2\right]}$$

Now, $T^{-1}_{ij} = \frac{1}{m} \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{r^2} & 0 \\ 0 & 0 & \frac{1}{r^2\sin^2\theta} \end{pmatrix}$

Looking back at the rotating frame Lagrangian:

$$\mathcal{L}_{rotational} = \frac{m}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2 + r^2\sin^2\theta\,\dot{\phi}'^2\right) + m\Omega r^2 \sin^2\theta\,\dot{\phi}' + \frac{m}{2}\Omega^2 r^2 \sin^2\theta - V(r)$$

Comparing with general Lagrangian:

$$\boxed{\mathcal{L} = \frac{1}{2} T_{ij} \dot{q}_i \dot{q}_j + a_i \dot{q}_i - \mathcal{L}_0(q)}$$

We see, $a_i \dot{q}_i = m\Omega r^2 \sin^2\theta$

$\Rightarrow a_{\phi}(r, \theta, \phi) = m\Omega r^2 \sin^2\theta$

$\boxed{a_r = 0}$, $\boxed{a_\theta = 0}$, $\boxed{a_\phi = m\Omega r^2 \sin^2\theta}$

Extra Notes: __/__/____

So solving, $T_{ij}^{-1} = \frac{1}{m} \text{diag}\left(1, 1/r^2, \frac{1}{r^2 \sin^2\theta}\right)$

Radial Velocity:

$$\dot{r} = (T^{-1})_{rr}(P_r - a_r) = \frac{1}{m}(P_r - 0) = \frac{P_r}{m}$$

Polar Velocity:

$$\dot{\theta} = (T^{-1})_{\theta\theta}(P_\theta - a_\theta) = \frac{1}{mr^2}(P_\theta - 0) = \frac{P_\theta}{mr^2}$$

Azimuthal Velocity:

$$\dot{\phi} = (T^{-1})_{\phi\phi}(P_\phi - a_\phi) =$$

$$= \frac{1}{mr^2 \sin^2\theta}(P_\phi - m\Omega r^2 \sin^2\theta)$$

$$= \frac{P_\phi}{mr^2 \sin^2\theta} - \Omega .$$

Hamiltonian in Matrix Form:

$$H = \frac{1}{2} T_{ij}^{-1} [P_i - a_i][P_j - a_j] - L_0$$

Writing it in Matrix form:

$$\boxed{H = \frac{1}{2}(\mathbb{P} - \mathbb{a})^T \mathbb{T}^{-1} (\mathbb{P} - \mathbb{a}) - L_0}$$

Here, $(\mathbb{P} - \mathbb{a}) = \begin{pmatrix} P_r - a_r \\ P_\theta - a_\theta \\ P_\phi - a_\phi \end{pmatrix} = \begin{pmatrix} P_r \\ P_\theta \\ P_\phi - m\Omega r^2 \sin^2\theta \end{pmatrix}$

[P. T. O.]

$$\therefore H = \frac{1}{2}(\mathbb{P}-\mathbb{a})^T \mathbb{I}^{-1}(\mathbb{P}-\mathbb{a}) - L_0(q)$$

$$= \frac{1}{2}\begin{pmatrix} p_r \\ p_\theta \\ p_\phi - m\Omega r^2\sin^2\theta \end{pmatrix} \frac{1}{m}\begin{pmatrix} 1 & 0 & 0 \\ 0 & r^2 & 0 \\ 0 & 0 & r^2\sin^2\theta \end{pmatrix}\begin{pmatrix} p_r \\ p_\theta \\ p_\phi - m\Omega r^2\sin^2\theta \end{pmatrix} - L_0(q)$$

Expanding this we get:

$$H = \frac{1}{2m}p_r^2 + \frac{1}{2mr^2}p_\theta^2 + \frac{(p_\phi - m\Omega r^2\sin^2\theta)^2}{2mr^2\sin^2\theta} - L_0(q)$$

Substituting: $L_0 = \frac{1}{2}m\Omega^2 r^2\sin^2\theta - V(r)$

$$H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{(p_\phi - m\Omega r^2\sin^2\theta)^2}{2mr^2\sin^2\theta} - \frac{1}{2}m\Omega^2 r^2\sin^2\theta + V(r)$$

$$\Rightarrow \boxed{H = \frac{p_r^2}{2m} + \frac{p_\theta^2}{2mr^2} + \frac{p_\phi^2}{2mr^2\sin^2\theta} - \Omega p_\phi + V(r)}$$

Symplectic Geometry:

$L(q, \dot{q}, t) \longrightarrow H(q, p, t) \rightarrow$ In 2N configuration space:

$\Gamma = (q_1, \dots, q_N, p_1, \dots, p_N) \in \mathbb{R}^{2N}$

[Phase space, has a natural geometric structure]

We are going to construct a geometry with: $\{q, p\}$: $\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}\ ;\ \dot{p}_i = -\frac{\partial H}{\partial q_i}}$

The phase space is 2N-dimensional with coordinates:

$$(q_1, \ldots, q_N, p_1, \ldots, p_N)$$

We relabel these as a single vector $\eta \in \mathbb{R}^{2N}$ for convenience:

$$\boxed{\eta_i = \begin{cases} q_i, & i = 1, \ldots, N \\ p_{i-N}, & i = N+1, \ldots, 2N \end{cases}}$$

Or equivalently,

$$\eta = (q_1, \ldots, q_N, p_1, \ldots, p_N)^T$$

So, $\eta_1 = q_1, \ldots, \eta_N = q_N$

$\eta_{N+1} = p_1, \ldots, \eta_{2N} = p_N$

- Symplectic Structure in η notation:

The Poisson Bracket can be written compactly using the Symplectic Matrix J.

$$\mathbb{J} = \begin{pmatrix} \phi_N & \mathbb{I}_N \\ -\mathbb{I}_N & \phi_N \end{pmatrix}_{2N \times 2N}$$

$\phi = $ null matrix $_{N \times N}$

$\mathbb{I}_N = $ Identity matrix of $N \times N$

$$\boxed{\begin{aligned} \tilde{\mathbb{J}} &= \mathbb{J}^{-1} = -\mathbb{J} \\ |\mathbb{J}| &= +1 \Rightarrow \text{Important} \end{aligned}}$$

In component form: $\boxed{\dot{\eta}_i = \sum_{j=1}^{2N} J_{ij} \frac{\partial H}{\partial \eta_j}}$

$\boxed{\text{Complete representation of HEOM}} \Rightarrow \left\{ \dot{q}_i = \frac{\partial H}{\partial p_i}, \ \dot{p} = -\frac{\partial H}{\partial q} \right.$

Canonical Transformation:

- **Canonical variables:** are pair of variables denoted as : (q_i, P_i), $i = 1, 2, \dots, n$, That describe the state of a Hamiltonian system in phase space and satisfy Hamilton's canonical equations of motion:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial P_i}} \quad , \quad \boxed{\dot{p} = -\frac{\partial H}{\partial q_i}}$$

They are called canonical because they obey the Canonical Poisson bracket relations:

$$\{q_i, q_j\} = 0 \,, \quad \{P_i, P_j\} = 0 \,, \quad \{q_i, P_i\} = \delta_{ij}$$

- **Canonical transformations:** Are transformations from one set of canonical ~~transformation~~ variables to another set of canonical variables.

$$\{q_i, P_i, H\} \longrightarrow \{Q_i, P_i, K\}$$

$$\dot{q}_i = \frac{\partial H}{\partial P_i} \,, \quad \dot{p}_i = -\frac{\partial H}{\partial q_i} \quad \Big| \quad \dot{Q}_i = \frac{\partial H}{\partial P_i} \,, \quad \dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

We have to make sure that the coordinates Q_i's are cyclic and corresponding momenta P_i's are conserved.

- We want the form of action to remain same:

$$\boxed{\delta I = \delta \int [p_i \dot{q}_i - H] dt = \delta \int [P_i \dot{Q}_i - K] dt}$$

[*] In Hamiltonian mechanics, form of action is: $I = \int_{t_1}^{t_2} \left(\sum_i p_i \dot{q}_i - H(q_i, p_i, t)\right) dt$

This equality means the form of Hamilton's equations is preserved, hence transformation is canonical.

- <u>Generating Function of 1st Type:</u>

Let us consider, $F = F_i(q_i, Q_i, t)$

Condition for a canonical transformation is that the action integrand changes only by a total time derivative:

$$\boxed{p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{dF_i}{dt}}$$

[Because variation $\delta \int_{t_1}^{t_2} \frac{dF}{dt} dt = 0$, it does not affect eqn of motion]

Now, $$\boxed{\frac{dF_i}{dt} = \sum_i \frac{\partial F_i}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial F_i}{\partial Q_i} \dot{Q}_i + \frac{\partial F_i}{\partial t}}$$

Substituting this, $p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_i}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial Q_i} \dot{Q}_i + \frac{\partial F_i}{\partial t}$

Collecting the coefficients of the independent time derivatives $\dot{q}_i$ and $\dot{Q}_i$, we get the following relations:

$$\boxed{p_i = \frac{\partial F_1}{\partial q_i}} \quad \boxed{P_i = -\frac{\partial F_1}{\partial Q_i}} \quad \boxed{K = H + \frac{\partial F_1}{\partial t}}$$

→ Transformation equations generated by $F_1(q_i, Q_i, t)$.

2nd Type of generating Function $F_2(q, P, t)$

$$F_2 = F_2(q, P, t)$$

$$p_i \dot{q}_i - H(q, p, t) = P_i \dot{Q}_i - K(Q, P, t) + \frac{d}{dt} F_2(q, P, t)$$

$$\frac{dF_2}{dt} = \sum_i \frac{\partial F_2}{\partial q_i} \dot{q}_i + \sum_i \frac{\partial F_2}{\partial Q_i} \dot{Q}_i + \frac{\partial F_2}{\partial t}$$

Substituting this into the identity:

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_2}{\partial q_i} \dot{q}_i + \frac{\partial F_2}{\partial P_i} \dot{P}_i + \frac{\partial F_2}{\partial t}$$

So, we get type-2 transformations:

$$\boxed{p_i = \frac{\partial F_2}{\partial q_i}} \quad \boxed{Q_i = \frac{\partial F_2}{\partial P_i}} \quad \boxed{K = H + \frac{\partial F_2}{\partial t}}$$

type 2 generating function $F_2(q, P, t)$, it is related to the type-1 function $F_1(q, Q, t)$ by the Legendre transform:

$$F_2(q, P, t) = F_1(q, Q, t) + \sum_i P_i Q_i$$

here, Q_i is an implicit function of (q, P, t).

One may start from the canonical identity:

$$p_i \dot{q}_i - H(q,p,t) = P_i \dot{Q}_i - K(Q,P,t) + \frac{d}{dt}F_1(q,Q,t)$$

Replacing F_1 by $F_2 - \sum_i P_i Q_i$

$$\frac{dF_1}{dt} = \frac{d}{dt}\left(F_2(q,P,t) - \sum_i P_i Q_i\right)$$

$$\boxed{\frac{dF_1}{dt} = \frac{dF_2}{dt} - \sum_i \left(\dot{P}_i Q_i + P_i \dot{Q}_i\right)}$$

- Substituting into canonical identity:

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{dF_2}{dt} - \sum_i \left(\dot{P}_i Q_i + P_i \dot{Q}_i\right)$$

$$\Rightarrow \boxed{p_i \dot{q}_i - H = -K + \frac{dF_2}{dt} - \sum_i \dot{P}_i Q_i}$$

Now,

$$\frac{dF_2}{dt} = \sum_i \frac{\partial F_2}{\partial q_i}\dot{q}_i + \sum_i \left(\frac{\partial F_2}{\partial P_i}\right)\dot{P}_i + \frac{\partial F_2}{\partial t} - \sum_i \dot{P}_i Q_i$$

Now we group terms containing $\dot{q}_i$ and $\dot{P}_i$:

$$p_i \dot{q}_i - H = -K + \sum_i \frac{\partial F_2}{\partial q_i}\dot{q}_i + \sum_i \left(\frac{\partial F_2}{\partial P_i} - Q_i\right)\dot{P}_i + \frac{\partial F_2}{\partial t}$$

- Coefficient of $\dot{q}_i$: $\boxed{p_i = \frac{\partial F_2}{\partial q_i}}$

- Coefficient of $\dot{P}_i$:

$$0 = \frac{\partial F_2}{\partial P_i} - Q_i \Rightarrow \boxed{Q_i = \frac{\partial F_2}{\partial P_i}}$$

- $K = H + \frac{\partial F_2}{\partial t}$

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Generating Function of 3-rd type:

$$F_3 = F_3(p, Q, t) = F_1(q, Q, t) - \sum_i q_i p_i$$

Here, q_i is an implicit functions of (p, Q, t)

Relation of action:

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{d}{dt}(F_3(p, Q, t) + q_i p_i)$$

Expanding the derivative:

$$\frac{d}{dt}(F_3 + q_i p_i) = \frac{\partial F_3}{\partial p_i}\dot{p}_i + \frac{\partial F_3}{\partial Q_i}\dot{Q}_i + \frac{\partial F_3}{\partial t} + \dot{q}_i p_i + q_i \dot{p}_i$$

$$\Rightarrow \boxed{q_i = -\frac{\partial F_3}{\partial p_i}} \quad \boxed{P_i = -\frac{\partial F_3}{\partial Q_i}} \quad \boxed{K = H + \frac{\partial F_3}{\partial t}}$$

Generating Function of 4-th type:

$$F_4(p, P, t) = F_2(q, P, t) - \sum_i q_i p_i$$

(Equivalently) $F_4(p, P, t) = F_3(p, Q, t) - \sum_i P_i Q_i$

One may write the action identity as:

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{d}{dt}(F_4(p, P, t) + p_i q_i - Q_i P_i)$$

$$\boxed{q_i = -\frac{\partial F_4}{\partial p_i}} \quad \boxed{Q_i = \frac{\partial F_4}{\partial P_i}} \quad \boxed{K = H + \frac{\partial F_4}{\partial t}}$$

For 2-D systems, we can construct a generating function that depends on a mixed set of old and new coordinates/momenta:

$$F = F(q_1, P_2, P_1, Q_2, t)$$

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We define generating function,

- Type 1 :

$$F = F'(q_1, P_2, P_1, Q_2, t) + q_i p_i - P_i Q_i$$

$$\boxed{p_1 = \frac{\partial F'}{\partial q_1}} \quad \boxed{Q_1 = \frac{\partial F'}{\partial P_1}} \quad \boxed{q_2 = -\frac{\partial F'}{\partial p_2}}, \quad \boxed{P_2 = -\frac{\partial F'}{\partial Q_2}}$$

$$K = H + \frac{\partial F'}{\partial t}$$

- Type 2 : $F_2 = q_i P_i$

- Rules give : $p_j = \frac{\partial F_2}{\partial q_j} = P_j$, $Q_j = \frac{\partial F_2}{\partial P_j} = q_j$, $K = H$

$\because$ There is no time-dependence, hence, $K = H$.

- Type 3 : $F_3 = P_i Q_i$

Similarly, one would compute the new variable is:

$$q_i = -\frac{\partial F_3}{\partial p_i}, \quad P_i = -\frac{\partial F_3}{\partial Q_i}, \quad K = H$$

Let us consider the type-2 generating function:

$$F_2 = f_i(q_1, q_2, \ldots, q_n, t) P_j$$

- $$Q_i = \frac{\partial F_2}{\partial P_i} = \delta_{ij} f_i(q_1, q_2, \ldots, q_n, t)$$

$$\dot{Q}_i = \frac{\partial f_i}{\partial q_j} \dot{q}_j, \quad \cancel{P_i = \frac{\partial f_i}{\partial q_j}} \quad p_i = \frac{\partial f_i}{\partial q_j} P_j$$

- Let us make a change here, we add another variable:

$$F_2 = f_j(q_1, q_2, \ldots, q_n, t) P_j + g(q_1, q_2, \ldots, q_n, t)$$

$\Rightarrow$ New coordinates : $Q_i = \frac{\partial F_2}{\partial P_i} = f_i(q_1, q_2, \ldots, q_n, t)$

The addition of $g(q, t)$ does not affect Q_i, because g does not depend on P_i

- $p_i = \frac{\partial F_2}{\partial q_i} = \sum_j \frac{\partial f_j}{\partial q_i} P_j + \frac{\partial g}{\partial q_i}$

- $K = H + \sum_j \frac{\partial f_j}{\partial t} P_j + \frac{\partial g}{\partial t}$

- Another trivial transformation

$$F_1(q_1, q_2, \ldots, q_n, t) = \sum_{k=1}^{n} q_k Q_k$$

No explicit time dependence, so, $K = H$.

- Type 1 generating functions:

Canonical transformation equations are:

$$p_k = \frac{\partial F_1}{\partial q_k} \quad P_k = -\frac{\partial F_1}{\partial Q_k} \quad K = H + \frac{\partial F_1}{\partial t}$$

$\because F_1$ is not explicitly time-dependent, $K = H$

So, ▸ old momenta:

$$p_k = \frac{\partial F_1}{\partial q_k} = \frac{\partial}{\partial q_k} \sum_j q_j Q_j = Q_k$$

▸ New momenta:

$$P_k = -\frac{\partial F_1}{\partial Q_k} = -\frac{\partial}{\partial Q_k} \sum_j q_j Q_j = -q_k$$

◦ Resulting canonical transformation:

Q_k = free coordinate, $P_k = -q_k$

$p_k = Q_k$, $K = H$.

Let ~~us~~ us consider the case of a 1-D S.H.O.

Here, $\omega^2 = k/m$

$$\boxed{H(q,p) = \frac{p^2}{2m} + \frac{1}{2}kq^2 = \frac{1}{2m}[p^2 + (m\omega q)^2],}$$

- The goal: perform a canonical transformation $(q,p) \longrightarrow (Q,P)$ such that Q is cyclic.
- P will be conserved and the solution will follow trivially.

- Let us define:

- $\boxed{p = f(P)\cos Q}$ - $\boxed{q = \frac{f(P)}{m\omega}\sin Q}$

$$p^2 + m^2\omega^2 q^2 = f(P)^2\cos^2 Q + f(P)^2 \sin^2 Q$$

$$\Rightarrow p^2 + m^2\omega^2 q^2 = f(P)^2$$

To make Hamiltonian linear in P, we choose

$$f(P) = \sqrt{2m\omega P} \Rightarrow \boxed{H = \omega P}$$

Now, from type-1 generating function $F_1(q,Q)$:

$$p = \frac{\partial F_1}{\partial q} \qquad P = -\frac{\partial F_1}{\partial Q}$$

- Compute the ratio:

$$\frac{p}{q} = \frac{f(P)\cos Q}{\left(\frac{f(P)}{m\omega}\right)\sin Q} = m\omega \frac{\cos Q}{\sin Q}$$

$$p = \frac{\partial F_1}{\partial q} = m\omega \frac{\cos Q}{\sin Q} q$$

$$\therefore F_1(q,Q) = \int \frac{\partial F_1}{\partial q} dq = \frac{m\omega}{2}\frac{\cos Q}{\sin Q} q^2 + \underbrace{g(Q)}_{\text{can be set to zero}}$$

Now, $P = -\frac{\partial F_1}{\partial Q}$

Computing derivative:

$$\boxed{\frac{\partial F_1}{\partial Q} = \frac{m\omega}{2} q^2 \frac{d}{dQ}\left(\frac{\cos Q}{\sin Q}\right) = -\frac{m\omega q^2}{2\sin^2 Q}}$$

$$P = \frac{m\omega q^2}{2\sin^2 Q}.$$

$$\Rightarrow \boxed{q = \sqrt{\frac{2P}{m\omega}} \sin Q}$$

Computing H in terms of P and Q:

$$p^2 + (m\omega q)^2 = 2Pm\omega \cos^2 Q + 2Pm\omega \sin^2 Q$$

$$\Rightarrow p^2 + (m\omega q)^2 = 2Pm\omega$$

$$\boxed{H = \frac{p^2 + (m\omega q)^2}{2m} = \omega P}$$

So, H depends only on P $\rightarrow$ Q is cyclic.

From Hamiltonian's Equations:

$$\boxed{\dot{Q} = \frac{\partial H}{\partial P} = \omega} \qquad \boxed{\dot{P} = -\frac{\partial H}{\partial Q} = 0}$$

$$\Rightarrow \boxed{Q(t) = \omega t + \alpha}$$

Plugging $Q(t)$ into the canonical transformation:

- $q(t) = \sqrt{\frac{2P}{m\omega}} \sin(\omega t + \alpha)$
- $p(t) = \sqrt{2Pm\omega} \cos(\omega t + \alpha)$

_/_/_

▫ Extra Notes:

Phase – space circle:

$$\frac{q^2}{A^2} + \frac{p^2}{(m\omega A)^2} = 1 \Rightarrow$$

$Q \equiv \omega t + \alpha$ is the angle along the circle.
$P \equiv \frac{1}{2} m \omega A^2$ is the area$/2\pi$ enclosed. It is a conserved quantity.

— end of Extra Notes. —

▫ Back to Class Notes:

We saw,
$$\boxed{q(t) = \sqrt{\frac{2E}{m\omega^2}} \sin(\omega t + \alpha)}$$
$$\boxed{p(t) = \sqrt{2mE} \cos(\omega t + \alpha)}$$
[Using $P = \frac{E}{\omega}$]

As both $q(t)$ and $p(t)$ both are periodic, bounded functions:

$$\boxed{|q(t)| \le \sqrt{\frac{2E}{m\omega^2}}}$$
~~$\dagger p(t)$~~
▫ $|p(t)| \le \sqrt{2mE}$
▫ $E \equiv$ constant.

▫ Trajectory in phase space is a circle of radius: $\sqrt{2mE}$ → stable oscillatory motion.

▫ Symplectic Geometry and Canonical Transformation:

We define the phase space vector as:

$$\eta = (\eta_1, \eta_2, \ldots, \eta_{2N}) = (q_1, \ldots, q_N, p_1, \ldots, p_N)$$

$N \equiv$ number of degrees of freedom.

Hamilton's equations can be written compactly as:

$$\boxed{\dot{\eta}_i = \sum_{j=1}^{2N} J_{ij} \frac{\partial H}{\partial \eta^j}}, \quad J = \begin{pmatrix} \phi & \mathbb{I} \\ -\mathbb{I} & \phi \end{pmatrix}_{N \times N}$$

$$[J^T = J^{-1} = -J]$$

We consider a change of variables:

$$(q_i, p_i) \Rightarrow (Q_i, P_i), \quad i = 1, 2, \ldots, N$$

In vector notation:

$$\eta \mapsto \tilde{\eta} = (Q_1, \ldots, Q_N, P_1, \ldots, P_N)$$

The new variables are canonical if they satisfy the same Hamiltonian structure:

$$\boxed{\dot{\tilde{\eta}}_i = \sum_{j=1}^{2N} \tilde{J}_{ij} \frac{\partial K}{\partial \tilde{\eta}^j}}, \quad \tilde{J} = J.$$

- Time-derivative of new coordinates:

$$\dot{Q}_i = \sum_j \frac{\partial Q_i}{\partial q_j} \dot{q}_j + \sum_j \frac{\partial Q_j}{\partial p_j} \dot{p}_j$$

$$\Rightarrow \boxed{\dot{Q}_i = \sum_j \frac{\partial Q_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \sum_j \frac{\partial Q_i}{\partial p_j} \frac{\partial H}{\partial q_j}}$$

Similarly,

$$\boxed{\dot{P}_i = \sum_j \frac{\partial P_i}{\partial q_j} \frac{\partial H}{\partial p_j} - \sum_j \frac{\partial P_i}{\partial p_j} \frac{\partial H}{\partial q_j}}$$

- Reciprocal Relations:

- $\left(\frac{\partial Q_i}{\partial q_j}\right)_{q,p} = \left(\frac{\partial p_i}{\partial P_j}\right)_{Q,P}$
- $\left(\frac{\partial Q_i}{\partial p_j}\right)_{q,p} = -\left(\frac{\partial q_j}{\partial P_i}\right)_{Q,P}$
- $\left(\frac{\partial P_i}{\partial q_j}\right)_{q,p} = -\left(\frac{\partial p_j}{\partial Q_i}\right)_{Q,P}$
- $\left(\frac{\partial P_i}{\partial p_j}\right)_{q,p} = \left(\frac{\partial q_j}{\partial Q_i}\right)_{P,Q}$

Let, $\eta = (q_1, \dots, q_N, p_1, \dots, p_N)$

$$\zeta_i = \zeta_i (Q_1, \dots, Q_N, P_1, \dots, P_N)$$

We assume a smooth coordinate transformation.

$$\eta_i = \eta_i(\zeta_1, \dots, \zeta_{2N}), \quad \zeta_i = \zeta_i(\eta_1, \dots, \eta_{2N})$$

We differentiate ζ_i with respect to time:

$$\boxed{\dot{\zeta}_i = \frac{\partial \zeta_i}{\partial \eta_j} \dot{\eta}_j = M_{ij} \dot{\eta}_j} \quad \text{— (A)}$$

We define the Jacobian Matrix of the transformation:

$$M_{ij} = \frac{\partial \zeta_i}{\partial \eta_j}$$

Now, Hamilton's equations in the η-coordinates:

$$\boxed{\dot{\eta}_i = J_{ij} \frac{\partial H}{\partial \eta_j}} \quad \text{— (B)}, \quad \tilde{J} = J^{-1} = -J$$

We substitute (B) into (A):

$$\boxed{\dot{\zeta}_i = M_{ij} J_{jk} \frac{\partial H}{\partial \eta_k}} \quad \text{— (*)}$$

$$\boxed{\dot{\zeta}_i = J_{ij} \frac{\partial H}{\partial \zeta_i} = J_i \frac{\partial H}{\partial \eta_k} \frac{\partial \eta_k}{\partial \zeta_j} = J_{ij} M^{-1}_{jk} \frac{\partial H}{\partial \eta_k}}$$

Using the chain rule:

$$\boxed{\frac{\partial H}{\partial \eta} = \frac{\partial H}{\partial \zeta} \frac{\partial \zeta}{\partial \eta} = \left(\frac{\partial \eta}{\partial \zeta}\right)^T \frac{\partial H}{\partial \zeta} = M^{-T} \frac{\partial H}{\partial \zeta}} \quad \text{— (C)}$$

Putting (C) → (*)

$$\boxed{\dot{\zeta}_i = M_{ij} J_{jk} (M^{-T})_{kl} \frac{\partial H}{\partial \zeta_l} = M_{ij} J_{jk} (M^{-1})_{lk} \frac{\partial H}{\partial \zeta_l}}$$

~~Der~~ Show that $\mathbb{J} = M \mathbb{J} M^T$:

HEOM in η- coordinates:

$$\boxed{\dot{\eta} = \mathbb{J} \frac{\partial H}{\partial \eta}}$$

Differentiating $\xi(\eta)$ in time:

$$\boxed{\dot{\xi} = M_{ij} \dot{\eta}_{j} = M \mathbb{J} \frac{\partial H}{\partial \eta}}$$

Using chain rule:

$$\boxed{\frac{\partial H}{\partial \eta} = \frac{\partial \xi^T}{\partial \eta} \frac{\partial H}{\partial \xi} = M^T \frac{\partial H}{\partial \xi}}$$

Thus, $\boxed{\dot{\xi} = M J M^T \frac{\partial H}{\partial \xi}}$ — (A)

For ξ to satisfy Hamilton's equations in the standard form:

$$\dot{\xi} = J \frac{\partial \tilde{H}}{\partial \xi} \quad \text{— (B)}$$

If H and $\tilde{H}$ same function expressed in new variables, or differs only by time derivative of a generating function. This yields the symplectic condition:

$$\mathbb{J} = M \mathbb{J} M^T$$

__/__/___

Poisson Bracket:

Poisson Bracket of any two functions u, v of canonical variables (q, p) is defined as:

$$[u, v]_{p,q} = \left[\frac{\partial u}{\partial q_i}\frac{\partial v}{\partial p_i} - \frac{\partial v}{\partial q_i}\frac{\partial u}{\partial p_i}\right].$$

In Symplectic Notation:

$$[u, v]_\eta = \frac{\partial u}{\partial \eta_i} J_{ij} \frac{\partial v}{\partial \eta_j} = \frac{\partial \tilde{u}}{\partial \eta} J \frac{\partial v}{\partial \eta}$$

Some Poisson Brackets:

- $[q_j, q_k]_{q,p} = 0 = [p_j, p_k]_{q,p}$
- $[q_j, p_k]_{p,q} = \frac{\partial q_j}{\partial q_i}\frac{\partial p_k}{\partial p_i} - \cancel{\frac{\partial p_k}{\partial q_i}}\frac{\partial q_j}{\partial p_i}$ $\to 0$

$$= \delta_{ji}\delta_{ki}$$

- $[q_j, p_k]_{p,q} = \delta_{jk} \implies -[p_j, q_k]_{p,q} = \delta_{jk}$

- Symplectic:

$$\eta = (\eta_1, \ldots, \eta_{2N}) = (q_1, \ldots, q_N, p_1, \ldots, p_N)$$

Poisson Bracket between components is written as:

$$[\eta_i, \eta_j] = \{\eta_i, \eta_j\} = J_{ij}$$

Performing a canonical transformation:

$$\eta \to \zeta \quad , \quad \zeta = \zeta(\eta)$$

with Jacobian matrix: $M_{ij} = \dfrac{\partial \zeta_i}{\partial \eta_j}$

If $[u, v]$ invariant under canonical transformation ___/___/___

means $\equiv$ $\boxed{[u, v]_\eta = \frac{\partial \tilde{u}}{\partial \eta} J \frac{\partial v}{\partial \eta}}$

For the transformation to be canonical, the fundamental Poisson bracket relations must be preserved;

Hence, $\boxed{[\zeta, \zeta]_\eta = J}$, $\boxed{[\eta, \eta]_\zeta = J}$

It means, new coordinates (Q_i, P_i) obey the same Poisson bracket relations as old ones (q_i, p_i)

$\{Q_i, Q_j\} = 0$, $\{P_i, P_j\} = 0$, $\{Q_i, P_j\} = \delta_{ij}$

Now, $\zeta = \zeta(\eta)$, $M_{ji} = \frac{\partial \zeta_j}{\partial \eta_i}$

For any scalar function: $v(\eta) = v(\zeta(\eta))$

Now, $\boxed{\frac{\partial v}{\partial \eta_i} = \frac{\partial v}{\partial \zeta_j} \frac{\partial \zeta_j}{\partial \eta_i}}$

$\Rightarrow \frac{\partial v}{\partial \eta_i} = M_{ji} \frac{\partial v}{\partial \zeta_j} = \tilde{M} \frac{\partial v}{\partial \zeta}$ $[M^T = \tilde{M} = M]$

Let $\zeta = \zeta(\eta)$, $M = \frac{\partial \zeta}{\partial \eta}$, $M^T = M^{-1} = \frac{\partial \eta}{\partial \zeta}$

For any scalar function,

$u(\eta) = u(\zeta(\eta))$

$\frac{\partial u}{\partial \eta} = M^T \frac{\partial u}{\partial \zeta}$, or equivalently, $\frac{\partial \tilde{u}}{\partial \eta} = \tilde{M} \frac{\partial \tilde{u}}{\partial \zeta}$

Now, HEOM in original coordinates are:

$$\boxed{\dot{\eta} = J \frac{\partial H}{\partial \eta}}$$

Transforming to new variables ζ:

$$\boxed{\dot{\zeta} = M\dot{\eta} = MJ \frac{\partial H}{\partial \eta} = MJM^T \frac{\partial H}{\partial \zeta}}$$

__/__/__

In matrix form, if we denote $\tilde{u} = \frac{\partial u}{\partial \zeta}$, this becomes:

$$\frac{\partial \tilde{u}}{\partial \eta} = \tilde{M} \frac{\partial u}{\partial \zeta}, \quad \dot{\zeta} = M J \tilde{M} \frac{\partial v}{\partial \zeta}$$

Derivation of $[u, v]_\zeta$:

$$\boxed{[u, v]_\zeta = \frac{\partial \tilde{u}}{\partial \zeta} M J \tilde{M} \frac{\partial v}{\partial \zeta}}$$

Let, $\zeta = \zeta(\eta)$; $M = \frac{\partial \zeta}{\partial \eta}$; $\tilde{M} = M^{-1} = \frac{\partial \eta}{\partial \zeta}$

$$\boxed{[u, v]_\eta = \frac{\partial u}{\partial \eta_j} J_{ij} \frac{\partial v}{\partial \eta_j}}$$

Now, $\frac{\partial u}{\partial \eta_i} = \frac{\partial u}{\partial \zeta_k} \frac{\partial \zeta_k}{\partial \eta_i} = M_{ki} \frac{\partial u}{\partial \zeta_k}$ — (A)

Similarly, $\boxed{\frac{\partial v}{\partial \eta_j} = M_{ij} \frac{\partial v}{\partial \zeta_i}}$ — (B)

We substitute (A) and (B) into $[u, v]_\eta$:

$$[u, v]_\eta = M_{ki} \frac{\partial u}{\partial \zeta_k} J_{ij} M_{ij} \frac{\partial v}{\partial \zeta_i} = \frac{\partial u}{\partial \zeta_k} M_{ki} J_{ij} M_{lj} \frac{\partial v}{\partial \zeta_l}$$

Hence,

$$\boxed{[u, v]_\zeta = \frac{\partial \tilde{u}}{\partial \zeta} M J M^T \frac{\partial v}{\partial \zeta}}$$

If the transformation is canonical:

$[u, v]_\zeta = [u, v]_\eta \Rightarrow$ Poisson bracket must remain invariant.

$$\therefore \boxed{[u, v]_\zeta = \frac{\partial \tilde{u}}{\partial \zeta_i} J_{ij} \frac{\partial v}{\partial \zeta_j} = \frac{\partial \tilde{u}}{\partial \eta_i} J_{ij} \frac{\partial v}{\partial \eta_i} = [u, v]_\eta}$$

Whether transformed coordinate also canonical or not, we use this property:

- if η is canonical and if $[\zeta, \zeta]_\eta = J$, then ζ is also canonical.

Properties of Poisson Bracket:

① $[u, u] = 0$ ② $[u, v] = -[v, u]$

③ $[au + bv, w] = a[u, w] + b[v, w]$

④ $[uv, w] = u[v, w] + [u, w]v$

⑤ $[u, [v, w]] + [v, [w, u]] + [w, [u, v]] = 0$

- Final goal is to find a constant of motion. Let $u = u(q, p, t)$

$$\frac{du}{dt} = \frac{\partial u}{\partial q_i}\dot{q}_i + \frac{\partial u}{\partial p_i}\dot{p}_i + \frac{\partial u}{\partial t}$$

$$\Rightarrow \frac{du}{dt} = \frac{\partial u}{\partial q_i}\frac{\partial H}{\partial p_i} - \frac{\partial u}{\partial p_i}\frac{\partial H}{\partial q_i} + \frac{\partial u}{\partial t}$$

$$\Rightarrow \frac{du}{dt} = [u, H] + \frac{du}{dt}$$

Constant of Motion:

A constant of motion is a function of the system's phase-space variables $F(q_1, q_2, \ldots, q_N, p_1, p_2, \ldots, p_N)$ that remains constant along the trajectory of motion: it does not change with time:

$$\frac{dF}{dt} = 0$$

In Hamiltonian mechanics:

$$[F, H] + \frac{\partial F}{\partial t} = 0$$

In general, we can write in symplectic geometry:

$$\dot{\eta}_i = [\eta_i, H]$$

- $$\frac{dH}{dt} = \cancel{[H,H]}^{0} + \frac{\partial H}{\partial t} \Rightarrow \boxed{\frac{dH}{dt} = \frac{\partial H}{\partial t}}$$

$\equiv$ If Hamiltonian is not explicitly dependent on time, it is a constant of motion.

$\Rightarrow$ If u is a constant of motion:

$$\boxed{\frac{du}{dt} = 0 = [u,H] + \frac{\partial u}{\partial t}}$$

- For conserved quantity; $[u,H] = -\frac{\partial u}{\partial t}$

- If a function is not explicitly a function of time, then it is constant of motion if $[u,H]=0$

- Let u, v be constants of motion:

$$\Rightarrow \boxed{\frac{\partial u}{\partial t} = \frac{\partial v}{\partial t} = 0} \Rightarrow \boxed{[u,H] = [v,H] = 0}$$

Considering poisson bracket: $[u,v]$,

$$\boxed{\frac{d}{dt}[u,v] = [[u,v],H] + \frac{\partial}{\partial t}[u,v]}$$

$\because$ u, v have no explicit time dependence:

$$\frac{\partial}{\partial t}[u,v] = 0 \Rightarrow \boxed{\frac{d[u,v]}{dt} = [[u,v],H]}$$

Using Jacobi's identity for Poisson brackets:

$$[u,[v,H]] + [v,[H,u]] + [H,[u,v]] = 0$$

[as $[u,H]=0$, $[v,H]=0$]

$$\boxed{\Rightarrow 0 + 0 + [H,[u,v]] = 0} \Rightarrow [u,v] \equiv \text{const. of motion}$$

Trivial example of constant of motion:

$H = \frac{p^2}{2m}$, p_x, p_y are constants of motion

$[p_y, H] = [p_x, H] = 0 \Rightarrow [p_x, p_y] = 0$

A Non-trivial Example:

$$H = \frac{p_x^2 + p_y^2 + p_z^2}{2m} + V(r), \quad r = \sqrt{x^2+y^2+z^2}$$

(This is a central force problem)

→ It is complicated because it has too many constants of motion, i.e., 3D - system has 6-phase-space variables (x, y, z, p_x, p_y, p_z).

We simplify it by introducing angular momentum vector: $\vec{L} = \vec{r} \times \vec{p}$

- We define:

$\boxed{L_x = yp_z - zp_y}$ $\boxed{L_y = zp_x - xp_z}$ $\boxed{L_z = xp_y - yp_x}$

$$[L_x, H] = [yp_z - zp_y, H] = [yp_z, H] - [zp_y, H]$$

$$\Rightarrow [L_x, H] = [y, H]p_z + y[p_z, H] - [z, H]p_y - z[p_y, H]$$

Now, we know:

$\boxed{[y, H] = \dot{y} = \frac{\partial H}{\partial p_y} = \frac{p_y}{m}}$, similarly, $\boxed{\dot{x} = \frac{p_x}{m}}$, $\boxed{\dot{z} = \frac{p_z}{m}}$

$$\boxed{[p_z, H] = \dot{p}_z = -\frac{\partial H}{\partial q_z} = -\frac{\partial V}{\partial z} = -\frac{\partial V}{\partial r}\frac{\partial r}{\partial z} = -\frac{z}{r}\frac{\partial V}{\partial r}}$$

Similarly, $\boxed{[p_x, H] = -\frac{x}{r}\frac{\partial V}{\partial r}}$; $\boxed{[p_y, H] = -\frac{y}{r}\frac{\partial V}{\partial r}}$

Now, $[L_x, H] = \frac{\partial L_x}{\partial y}\frac{\partial H}{\partial p_y} + \frac{\partial L_x}{\partial z}\frac{\partial H}{\partial p_z} - \frac{\partial L_x}{\partial p_y}\frac{\partial H}{\partial y} - \frac{\partial L_x}{\partial p_z}\frac{\partial H}{\partial z}$

$$= \frac{p_z p_y}{m} + (-p_y)\frac{p_z}{m} - (-z)\frac{y}{r}\frac{\partial V}{\partial r} - y\frac{z}{r}\frac{\partial V}{\partial r}$$

$$= \frac{p_z p_y}{m} - \frac{p_y p_z}{m} + \frac{zy}{r}\frac{\partial V}{\partial r} - \frac{yz}{r}\frac{\partial V}{\partial r}$$

$\Rightarrow \boxed{[L_x, H] = 0}$. Similarly, $\boxed{[L_y, H] = [L_z, H] = 0}$

- Show that $[L_x, L_y] = L_z$

- For L_x: $\frac{\partial L_x}{\partial x} = 0$ $\quad \frac{\partial L_x}{\partial y} = p_z$ $\quad \frac{\partial L_x}{\partial z} = -p_y$

$\frac{\partial L_x}{\partial p_x} = 0$ $\quad \frac{\partial L_x}{\partial p_y} = -z$ $\quad \frac{\partial L_x}{\partial p_z} = y$

- For L_y: $\frac{\partial L_y}{\partial x} = -p_z$ $\quad \frac{\partial L_y}{\partial y} = 0$ $\quad \frac{\partial L_y}{\partial z} = p_x$

$\frac{\partial L_y}{\partial p_x} = z$ $\quad \frac{\partial L_y}{\partial p_y} = 0$ $\quad \frac{\partial L_y}{\partial p_z} = -x$

$$[L_x, L_y] = \sum_i \frac{\partial L_x}{\partial q_i}\frac{\partial L_y}{\partial p_i} - \frac{\partial L_x}{\partial p_i}\frac{\partial L_y}{\partial q_i}$$

$$= \frac{\partial L_x}{\partial x}\frac{\partial L_y}{\partial p_x} + \frac{\partial L_x}{\partial y}\frac{\partial L_y}{\partial p_y} + \frac{\partial L_x}{\partial z}\frac{\partial L_y}{\partial p_z}$$
$$- \left(\frac{\partial L_x}{\partial p_x}\frac{\partial L_y}{\partial x} + \frac{\partial L_x}{\partial p_y}\frac{\partial L_y}{\partial y} + \frac{\partial L_x}{\partial p_z}\frac{\partial L_y}{\partial z}\right)$$

$$= (0)(z) - (p_z)(0) + (-p_y)(-x)$$
$$- (0(-p_z) + (-z)(0) + (y)(p_x))$$

$\boxed{[L_x, L_y] = p_y x - y p_x = L_z}$ $\Rightarrow$ $\boxed{[L_x, L_y] = L_z}$

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We look at a canonical transformation:

$Q_i = q_i + \delta q_i$; $P_i = p_i + \delta p_i$

$$(q,p) \xrightarrow{(\delta p, \delta q)} (Q,P)$$

$$\boxed{\zeta = \eta_i + \delta\eta_i}$$

(δp_i) and (δq_i) are very small, hence 2nd order terms in $(\delta\eta_i)^2$ are ~~is~~ ignorable.

Let the generating function be:

$$\boxed{F = F_0(q,P,t) + \epsilon G(q,p,t)}, \quad \epsilon << 1$$

$$\Rightarrow F = q_i P_i + \epsilon G(q,p,t), \quad \epsilon << 1$$

$F_0 = q_i P_i$

So, the new canonical variables are:

$$Q_i = \frac{\partial F}{\partial P_i} = q_i + \epsilon \frac{\partial G}{\partial P_i}$$

$$P_i = \frac{\partial F}{\partial q_i} = p_i + \epsilon \frac{\partial G}{\partial q_i}$$

$$\delta p_i = P_i - p_i = -\epsilon \frac{\partial G}{\partial q_i}$$

$$\delta q_i = Q_i - q_i = -\epsilon \frac{\partial G}{\partial P_j}$$

Now, $\frac{\partial G}{\partial P_j} = \sum_k \frac{\partial G}{\partial p_k} \frac{\partial p_k}{\partial P_j}$

For infinitesimal translation:

$$\boxed{P_k = p_k + \epsilon \frac{\partial G}{\partial q_k}} \Rightarrow p_k = P_k - \epsilon \frac{\partial G}{\partial q_k}$$

Then the derivative of p_k w.r.t. P_j is:

$$\boxed{\frac{\partial p_k}{\partial P_j} = \frac{\partial}{\partial P_j}\left(P_k - \epsilon\frac{\partial G}{\partial q_k}\right) = \delta_{kj} - \epsilon\underbrace{\frac{\partial^2 G}{\partial q_k \partial P_j}}_{\text{Very small, as } \epsilon << 1}}$$

In matrix form, this is:

$$\frac{\partial \hat{p}}{\partial \hat{P}} = \mathbb{I} - \epsilon\Phi + \cancelto{0}{O(\epsilon^2)} \text{ [Too small]}$$

So, $$\epsilon\frac{\partial G}{\partial P_j} = \epsilon\sum_k \frac{\partial G}{\partial p_k}\frac{\partial p_k}{\partial P_j}$$

$$\Rightarrow \epsilon\frac{\partial G}{\partial P_j} = \epsilon\sum_k \frac{\partial G}{\partial p_k}(\delta_{kj} - \epsilon.0) = \epsilon\frac{\partial G}{\partial p_j}$$

$$\Rightarrow \boxed{\epsilon\frac{\partial G}{\partial P_j} \simeq \epsilon\frac{\partial G}{\partial p_j}} \longrightarrow \text{Important}$$

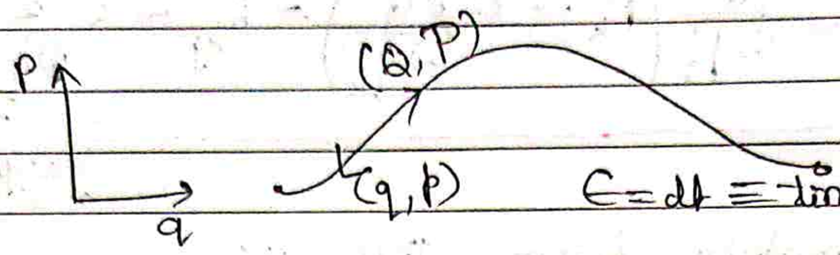
So, derivative w.r.t. new momentum reduces to derivative w.r.t. old momentum p_j upto 1st order in ϵ.

$$(q,p) \xrightarrow{\delta\eta} (Q,P)$$

- We can write: $\boxed{\delta\eta = \epsilon J\frac{\partial G}{\partial \eta}}$

$$\left[\text{Because: } \delta q_i = \epsilon\frac{\partial G}{\partial p_i}, \quad \delta p_i = -\epsilon\frac{\partial G}{\partial q_i}\right]$$

Suppose a system is evolving in time in phase space:



$\epsilon = dt \equiv$ time step.

$\xi(t) \rightarrow \xi(t_0 + dt) \Rightarrow$ Canonical transformation is to take a particle from 1 point in phase space to another in small time step dt.

- Old canonical variables:

$$\eta_j \, ; \, j = 1, \ldots, 2N$$

$$\eta = (\underbrace{q_1, q_2, \ldots, q_N}_{\text{positions}}, \underbrace{p_1, p_2, \ldots p_N}_{\text{momenta}})^T$$

- new variables: $\zeta_i(t+dt)$, $i = 1, \ldots, 2N$

$$\zeta(\eta, t+dt) = \eta_i + dt\,\dot{\eta}_i = \eta_i + dt\, J_{ik} \frac{\partial H}{\partial \eta_k}$$

We define the infinitesimal shift:

$$\boxed{\delta \eta_i = \zeta_i(\eta, t+dt) - \eta_i = dt\, J_{ik} \frac{\partial H}{\partial \eta_k}}$$

So, $$M_{ij} = \frac{\partial \zeta_i}{\partial \eta_j} = \frac{\partial}{\partial \eta_j}\left(\eta_i + dt\, J_{ik} \frac{\partial H}{\partial \eta_k}\right)$$

$$\Rightarrow \boxed{M_{ij} = \frac{\partial \zeta_i}{\partial \eta_j} = \delta_{ij} + dt\, J_{ik} \frac{\partial^2 H}{\partial \eta_k \partial \eta_j}}$$

In matrix notation: $\boxed{\hat{M} = \mathbb{I} + \epsilon \mathbb{J} \frac{\partial^2 G}{\partial \eta^2}}$

▫ Now, HEOM ~~equals~~ can be written as:

$$\dot{\eta} = J \frac{\partial H}{\partial \eta}$$

$$\boxed{\delta \eta = \frac{\eta(t) - \eta(t_0)}{dt} = J \frac{\partial H}{\partial \eta}}$$

When, $\zeta_i = \eta_i + \delta \eta_i$, $\delta \eta_i << 1$.

$$M_{ij} = \frac{\partial \zeta_i}{\partial \eta_j} = \frac{\partial(\eta_i + \delta \eta_i)}{\partial \eta_j} = \frac{\partial \eta_i}{\partial \eta_j} + \frac{\partial(\delta \eta_i)}{\partial \eta_j}$$

$\because$ $\frac{\partial \eta_i}{\partial \eta_j} = \delta_{ij}$

$$M_{ij} = \delta_{ij} + \frac{\partial(\delta \eta_i)}{\partial \eta_j}$$

$$\Rightarrow \boxed{M_{ij} = \delta_{ij} + \epsilon J_{ik} \frac{\partial^2 G}{\partial \eta_k \partial \eta_j}}$$

In matrix notation:

$$\boxed{M = \mathbb{I} + \epsilon \mathbb{J} \frac{\partial^2 G}{\partial \eta_i \partial \eta_j}}$$

In matrix notation:

$$[\eta, u] = \frac{\partial \tilde{\eta}}{\partial \eta} \mathbb{J} \frac{\partial u}{\partial \eta}$$

$$\Rightarrow \boxed{[\eta, H]_\eta = \dot{\eta} = J \frac{\partial H}{\partial \eta}}$$

$$\Delta \eta = dt\, \dot{\eta}$$

$$\Rightarrow \Delta \eta = dt\, J \frac{\partial H}{\partial \eta}$$

$$\Rightarrow \boxed{\Delta \eta = dt\, [\eta, H]}$$

Now, $\dot{\eta} = J \frac{\partial H}{\partial \eta}$

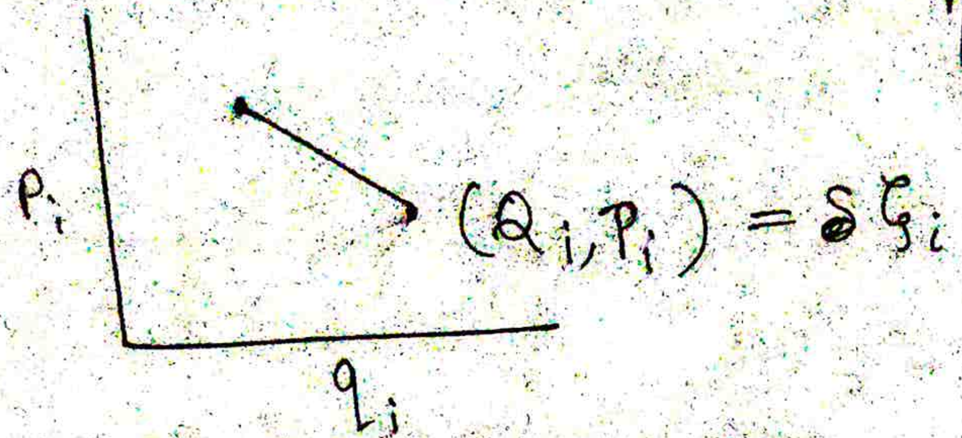
$$\Rightarrow \int_{t_0}^{t} \eta \, dt = \int_{t_0}^{t} J \frac{\partial H}{\partial \eta} dt'$$

$$\Rightarrow \eta(t) = \eta(t_0) + dt \, J \frac{\partial H}{\partial \eta}$$ [$dt = t - t_0$ very small]

Classical Mechanics:

Infinitesimal Canonical Transform:

$$\eta_i = \begin{pmatrix} q_i \\ p_i \end{pmatrix}$$



Here, transformation very small (infinitesimal).

$$\boxed{Q_i = q_i + \delta q_i} \quad \boxed{P_i = p_i + \delta p_i}$$

as δp and δq very small: $\Rightarrow \begin{pmatrix} \delta p \\ \delta q \end{pmatrix}^2$

$$\boxed{\delta\eta << 1}$$

- We are not doing it from geometrical point of view, rather from canonical variable's POV.

Symplectic Geometry:

$$\eta_i = \begin{pmatrix} q \\ p \end{pmatrix} ; \quad \zeta_i = \eta_i + \delta\eta_i$$

Degree of freedom only in terms with q, not in terms of p.
[2-D phase space $\Rightarrow$ 1 d.o.f.] [P.T.O.]

- In this case, we use generating function of Type 2 : $\boxed{F_2 = F_2(q, P, t)}$

$$\Rightarrow \boxed{F_2 = q_i P_i + \epsilon G(q, P, t)}$$

$$\boxed{p_i = \frac{\partial F_2}{\partial q_i}} \quad \text{—(i)} \qquad \boxed{Q_i = \frac{\partial F_2}{\partial P_i}} \quad \text{—(ii)}$$

- Standard Relation that relate the generating Function with the old and the new canonical variables.

$$p_i = P_i + \epsilon \frac{\partial G}{\partial q_i} \quad \text{[From (i)]}$$

$$\Rightarrow P_i = p_i - \epsilon \frac{\partial G}{\partial q_i}$$

$$\Rightarrow P_i - p_i = -\epsilon \frac{\partial G}{\partial q_i}$$

$$\Rightarrow \boxed{\delta p_i = -\epsilon \frac{\partial G}{\partial q_i}}$$

$$Q_i = q_i + \epsilon \frac{\partial G}{\partial P_i}$$

It is not convenient to have a relation with the new coordinates involved (Here, P_i).

We utilize :

$P_i = p_i - \epsilon \frac{\partial G}{\partial q_i}$ to replace P_i with p_i.

$$\therefore Q_i = q_i + \epsilon \frac{\partial G}{\partial P_i}$$

$$\Rightarrow Q_i = q_i + \epsilon \frac{\partial G}{\partial p_j} \frac{\partial p_j}{\partial P_i}$$

as $p_j = P_j + \epsilon \frac{\partial G}{\partial q_j}$

$$\Rightarrow \boxed{\frac{\partial p_j}{\partial P_i} = \delta_{ij} + \epsilon \frac{\partial^2 G}{\partial q_j \partial P_i}}$$

Writing clearly,

$$\boxed{\frac{\partial p_j}{\partial P_i} = \delta_{ij} + \epsilon \frac{\partial^2 G}{\partial q_j \partial P_i}}$$

Hence, $Q_i = q_i + \epsilon \dfrac{\partial G}{\partial p_j} \dfrac{\partial p_j}{\partial P_i}$

$$\Rightarrow Q_i = q_i + \epsilon \frac{\partial G}{\partial p_j} (\delta_{ij} + \epsilon [\ ])$$

$$\Rightarrow \boxed{Q_i = q_i + \epsilon \frac{\partial G}{\partial p_j} \left(\delta_{ij} + \epsilon \frac{\partial^2 G}{\partial q_j \partial P_i} \right)}$$

- We do not take into account the term $\epsilon \dfrac{\partial^2 G}{\partial q_j \partial P_i}$, [$\partial q_j \, \partial P_i$ very small]

$$\Rightarrow Q_i = q_i + \delta_{ij} \epsilon \frac{\partial G}{\partial p_j}$$

$$\Rightarrow Q_i = q_i + \epsilon \frac{\partial G}{\partial p_i}$$

$$\Rightarrow Q_i - q_i = \epsilon \frac{\partial G}{\partial p_i}$$

$$\Rightarrow \boxed{\delta q_i = \epsilon \frac{\partial G}{\partial p_i}}$$

also, $\boxed{\delta p_i = -\epsilon \dfrac{\partial G}{\partial q_i}}$

Index notation

- We have: $\zeta_i = \eta_i + \delta \eta_i$, $\boxed{\delta \eta_i = J_{ij} \epsilon \dfrac{\partial G}{\partial \eta_j}}$

Matrix Notation $\rightarrow$ $\boxed{\delta \eta = J \epsilon \dfrac{\partial G}{\partial \eta}}$

Poisson Bracket:

$$[u, v]_\eta = \frac{\partial \tilde{u}}{\partial \eta} J \frac{\partial v}{\partial \eta} \quad \text{[Definition]}$$

Let, $u = \eta$, $v = u$ [u and u not same]

$$\cancel{[\zeta, u]} \quad [\eta, u]_\eta = \mathbb{I} . J . \frac{\partial u}{\partial \eta} = J \frac{\partial u}{\partial \eta}$$

So, $\delta \eta_i = J_{ij} \epsilon \frac{\partial G}{\partial \eta_j}$

$$\boxed{[\eta_i, u] = J \frac{\partial u}{\partial \eta_i}}$$

$$\Rightarrow \delta \eta_i = \epsilon J \frac{\partial G}{\partial \eta}$$

$$\Rightarrow \boxed{\delta \eta_i = \epsilon [\eta_i, G]_\eta}$$

- Poisson Bracket can be written in terms of a transformation Matrix.

$G(\eta, t)$, $\boxed{\delta \eta_i = \epsilon [\eta_i, G]}$

$$\zeta_i = \eta_i + \delta \eta_i$$

Now, $\boxed{\dot{\eta} = J_{ij} \frac{\partial H}{\partial \eta_j}} = [\eta, H]$

Now, $\boxed{[\eta, u] = \frac{\partial u}{\partial \eta}}$ $\boxed{\dot{\eta}_i = [\eta_i, H]}$

$$\boxed{\frac{\eta_i(dt+t) - \eta_i(t)}{dt} = [\eta_i, H]}$$

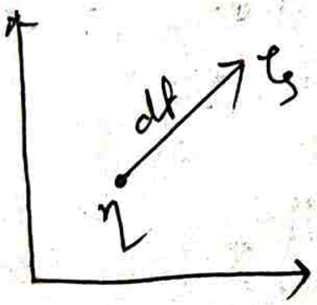
↳ By 1^{st} principle.

$$\boxed{\eta_i(t+dt) = \eta_i(t) + dt\,[\eta_i, H]}$$

$\eta_i(t)$, $\quad \zeta_i = \eta_i(\cancel{t}\, t + dt)$

$$\Rightarrow \boxed{\zeta_i = \eta_i + dt\,[\eta_i, H]}$$

Hamiltonian is a time translation operator.



- The small change in position is given by:

$$\eta + \underbrace{dt[\eta, H]}_{\text{This quantity}}$$

- $\boxed{\text{If all the coordinates are cyclic, life is very comfortable ☺.}}$

$M_{ij} \equiv$ jacobian matrix $= \dfrac{\partial \zeta_i}{\partial \eta_j}$ [For, $\zeta_i = \zeta_i(\eta)$]

- $\boxed{\text{How to find out if a particular transformation is canonical or not? } \underline{\text{Ans:}} \text{ If } MJ\tilde{M} = J}$

$$\boxed{\zeta_i = \eta_i + J_{ij}\epsilon \frac{\partial G}{\partial \eta_j}}$$

Now we construct the M_{ij} matrix:

$$\boxed{M_{ij} = \delta_{ij} + \epsilon J_{ik} \frac{\partial^2 G}{\partial \eta_j \partial \eta_k}}$$

$$\boxed{M_{ij} = \delta_{ij} + \epsilon J_{ik} \frac{\partial^2 G}{\partial \eta_j \partial \eta_k}}$$

↳ Find out, is this identity trivially coming out from $MJ\tilde{M} = J$ identity?

- $$\mathcal{L}_0 = \frac{m}{2}\left(\frac{1}{\sigma^2 + \tau^2}\right)(\dot{\sigma}^2 + \dot{\tau}^2)$$

$$\frac{\partial \mathcal{L}_0}{\partial \dot{\sigma}} = \frac{m}{2}\left(\frac{1}{\sigma^2 + \tau^2}\right) 2\dot{\sigma}$$

- $$\frac{\partial \mathcal{L}_0}{\partial \dot{\sigma}} = \frac{m\dot{\sigma}}{\sigma^2 + \tau^2} = p_\sigma$$

- $$\frac{d}{dt}\frac{\partial \mathcal{L}_0}{\partial \dot{\tau}} = \frac{m\dot{\tau}}{\sigma^2 + \tau^2} = p_\tau$$

$$\therefore \dot{\sigma} = \frac{p_\sigma(\sigma^2 + \tau^2)}{m}$$

$$\dot{\tau} = \frac{p_\tau(\sigma^2 + \tau^2)}{m}$$

$$H = \sum_i p_i \dot{q}_i - L = p_\sigma \dot{\sigma} + p_\tau \dot{\tau} - L$$

$$\Rightarrow H = p_\sigma\left(\frac{p_\sigma(\sigma^2 + \tau^2)}{m}\right) + p_\tau\left(\frac{p_\tau(\sigma^2 + \tau^2)}{m}\right) - L$$

$$\Rightarrow H = \frac{P_\sigma^2(\sigma^2+\tau^2)}{m} + \frac{P_\tau^2(\sigma^2+\tau^2)}{m} - \frac{m}{2}\left(\frac{1}{\sigma^2+\tau^2}\right)\left[\left(\frac{P_\sigma^2(\sigma^2+\tau^2)^2}{m^2}\right)+\left(\frac{p_\tau^2(\sigma^2+\tau^2)^2}{m^2}\right)\right]$$

$$\boxed{\dot{\sigma} = \frac{\partial H}{\partial P_\sigma}}, \quad \boxed{\dot{P}_\sigma = -\frac{\partial H}{\partial \sigma}} \quad [H.E.O.M]$$

Q.2) Let a q transformation be $q \to Q$, $p \to P$.

$$\left.\begin{aligned} Q &= q\cos\alpha - p\sin\alpha \\ P &= q\sin\alpha + p\cos\alpha \end{aligned}\right] \to \text{Rotatory transformation.}$$

$$\left.\begin{aligned} q &= Q\cos\alpha + P\sin\alpha \\ p &= -Q\sin\alpha + P\cos\alpha \end{aligned}\right] \to$$ Algebra করে বের করতে হবে LMAO.

$$\boxed{M = \begin{pmatrix} \frac{\partial Q}{\partial q} & \frac{\partial Q}{\partial p} \\ \frac{\partial P}{\partial q} & \frac{\partial P}{\partial p} \end{pmatrix}}$$

[We inverte the matrix $\begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} = M$

$\Rightarrow \det|M| = 1$

$$M = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\Big(\quad \Big) \quad M J \tilde{M} = J$$

Classical Mechanics :

- We use conservative system : $\vec{F} = -\nabla V$.
  Time independent system.
  T is purely quadratic in $\dot{q}_i \Rightarrow T = \frac{1}{2} m_{ij} \dot{q}_i \dot{q}_j$

$$T = \frac{1}{2} m \dot{q}^2$$

(*) We study small motion near equilibrium point.

(*) We start with 1-dimensional case :

$$L = T - V$$

(*) Potential $V(q) \Rightarrow L = \frac{1}{2} m \dot{q}^2 - V(q)$

- LEOM : $\boxed{m\ddot{q} = -\frac{\partial V}{\partial q}}$

The equilibrium condition is given by $\frac{\partial V}{\partial q} = 0$

Let equilibrium be at $q = q_0 \Rightarrow \left(\frac{\partial V}{\partial q}\right)_{q=q_0} = 0$

- $q = q_0 + \eta$, $\eta << 1$, $O(\eta^2) = 0$, $\eta^2 \simeq 0$

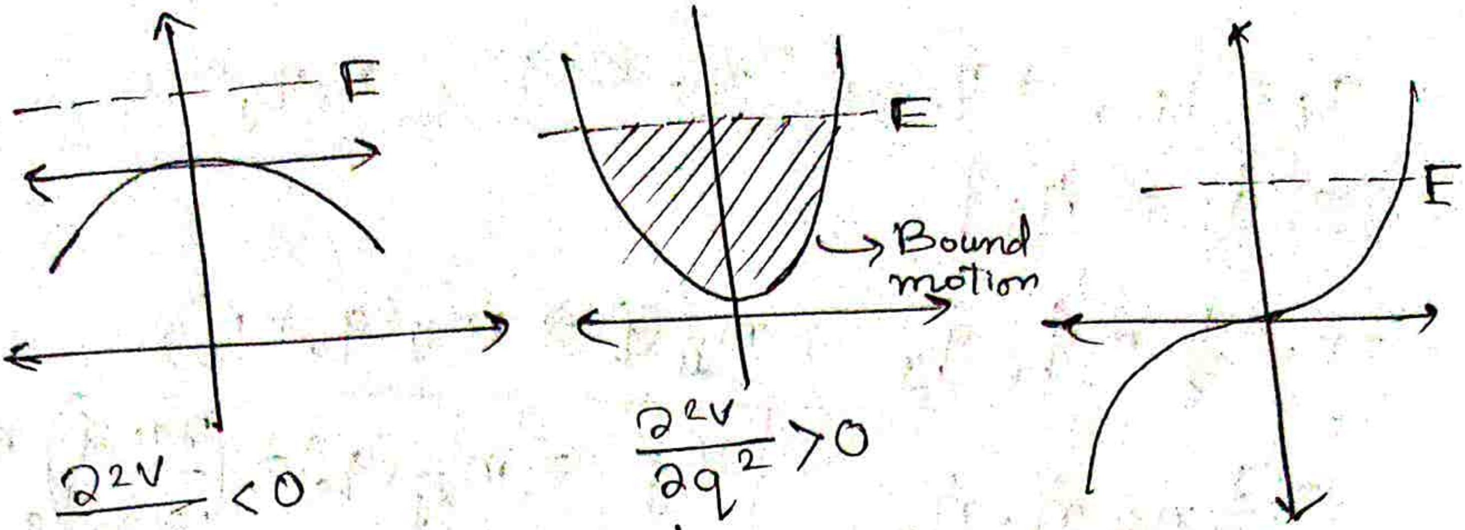
$$\dot{q} = \dot{\eta}, \quad V(q) = V(q_0 + \eta) = \underset{0}{\underset{\shortparallel}{V(q_0)}} + \left(\frac{\partial V}{\partial q}\right)_{q=q_0} \eta + \left(\frac{\partial^2 V}{\partial q^2}\right)_{q=q_0} \eta^2$$

- We assume;

$V(q_0) = 0$

$\left(\frac{\partial V}{\partial q}\right)_{q=q_0} = 0 \rightarrow$ at equilibrium

$$T = \frac{1}{2} m \dot{q}^2 = \frac{1}{2} m \dot{\eta}^2 \text{ [Lowest order term in } \eta]$$



$\frac{\partial^2 V}{\partial q^2} < 0$ $\Downarrow$ Unstable equilibrium

$\frac{\partial^2 V}{\partial q^2} > 0$ $\downarrow$ Stable equilibrium

$\frac{\partial^2 V}{\partial q^2} = 0$ $\downarrow$ Unstable.

$\boxed{\dot{r} = (E - V)^{1/2}}$ [$\boxed{E > V}$ for meaningful motion]

We are interested in small / bound motion around equilibrium point: $\boxed{\frac{\partial^2 V}{\partial q^2} > 0} \Rightarrow \boxed{\frac{\partial^2 V}{\partial q^2} = k^2}$

- $V(q) = \frac{1}{2} k^2 \eta^2$, $L = \frac{1}{2} m \dot{\eta}^2 - \frac{1}{2} k^2 \eta^2$

Note that our generalised coordinate for motion about equilibrium point is η.

$\ddot{\eta} + \frac{k^2}{m} \eta = 0$

$\eta = (\ \ ) \sin \omega t + (\ \ ) \cos \omega t$, $\omega^2 = \frac{k^2}{m}$

$\eta = Ce^{i\omega t} = Ce^{i\sqrt{\frac{k^2}{m}}\, t}$

System with 'n' degrees of freedom, ~~q_i's~~ q_i's generalized coordinates. Let $q_i(0)$ is stable equilibrium point.

[P.T.O.]

$q_i = q_{i(0)} + \eta_i$; $\eta_i << 1$, $\eta_i\eta_j \simeq 0$

$\Rightarrow \boxed{\dot{q}_i = \dot{\eta}_i}$

$T = \frac{1}{2} m \dot{q}_i \dot{q}_j$, $m_{ij}(q) \Rightarrow m_{ij}(q_0 + \eta)$

$= \frac{1}{2} m_{ij} \dot{\eta}_i \dot{\eta}_j$ $\quad \boxed{m_{ij} = m_{ij}(q_0) + \left(\frac{\partial m_{ij}}{\partial q_k}\right)_{q_0} \eta_k}$

$$\boxed{T = \frac{1}{2} m_{ij}(q_0)\dot{\eta}_i \dot{\eta}_j + O(\eta^3)}$$

$\boxed{M_{ij} \text{ is constant at } q = q_0.}$

$T = \frac{1}{2} T_{ij} \dot{\eta}_i \dot{\eta}_j$ are constants $T_{ij} = T_{ji}$

Symmetric matrix is +ve definite because K.E. is always constant.

- Physical systems with K.E. of the form $\boxed{T = \frac{1}{2} T_{ij} \dot{\eta}_i \dot{\eta}_j}$ for small motion, about an equilibrium point $q_i(0)$;

$\boxed{q_i = q_i(0) + \eta_i}$; $\eta_i << 1$

T_{ij} are constant T_{ij} is symmetric matrix ; $T_{ij} = T_{ji}$ $= \mathbb{T} = \tilde{\mathbb{T}}$. Let us expand the potential $V(q)$ around q_0:

$$\boxed{V(q) = V(q_0 + \eta) = V(q_0) + \left(\frac{\partial V}{\partial q_i}\right)_{q_0} \eta_i + \frac{1}{2}\left(\frac{\partial^2 V}{\partial q_i \partial q_j}\right)_{q_0} \eta_i \eta_j + O(\eta^3)}$$

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We are interested in motion around stable equilibrium pt q_0:

$$\boxed{V = \frac{1}{2} V_{ij} \eta_i \eta_j} \quad , \quad \boxed{V_{ij} = \left.\frac{\partial^2 V}{\partial q_i \partial q_j}\right|_{q_0} > 0,} \quad V_{ij} = V_{ji}$$

$$\boxed{L = \frac{1}{2} [T_{ij} \dot{\eta}_i \dot{\eta}_j - V_{ij} \eta_i \eta_j]}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\eta}_k}\right) - \frac{\partial L}{\partial \eta_k} = 0$$

$$\frac{\partial L}{\partial \dot{\eta}_k} = \frac{1}{2} T_{ij} (\delta_{ik} \dot{\eta}_j + \delta_{jk} \dot{\eta}_i)$$

$$\frac{\partial L}{\partial \dot{q}_k} = \frac{1}{2} (T_{kj} \dot{\eta}_j + T_{ik} \dot{\eta}_i)$$

$$\boxed{\frac{\partial L}{\partial \dot{q}_k} = \frac{1}{2} [T_{ki} \dot{\eta}_i + T_{ki} \dot{\eta}_i] = T_{ki} \dot{\eta}_i}$$

$$\boxed{\frac{\partial L}{\partial q_k} = - V_{ki} \eta_i}$$

$$\boxed{T_{ki} \ddot{\eta}_i + V_{ki} \eta_i = 0} \quad , \quad \boxed{\eta_i = a_i e^{-i\omega t}}$$

$$- T_{ki} a_i \omega^2 e^{-i\omega t} + V_{ki} a_i e^{-i\omega t} = 0$$

$$\Rightarrow V_{ki} a_i - \omega^2 T_{ki} a_i = 0$$

$$\Rightarrow V . A - \omega^2 \mathbb{T} A = 0 \Rightarrow (V - \omega^2 \mathbb{T}) A = 0$$

Classical Mechanics

Small Oscillations:

Physical systems with K.E. of the form:

$$\boxed{T = \frac{1}{2} T_{ij} \dot{\eta}_i \dot{\eta}_j}$$

For small motion about an equilibrium point

$$\boxed{q_{i(0)} : q_i = q_{i(0)} + \eta_i} \qquad \eta_j << 1$$

T_{ij} are constant $\Rightarrow$ T is a symmetric matrix.

$$\Rightarrow \boxed{T_{ij} = T_{ji}} \quad , \quad \boxed{T = \tilde{T}}$$

Let us expand the potential $V(q)$ around q_0:

$$V(q) = V(q_0 + \eta) = V(q_0) + \left(\frac{\partial V}{\partial q_i}\right)\Big|_{q_0} \eta_i$$

$$+ \frac{1}{2}\left(\frac{\partial^2 V}{\partial q_i \partial q_j}\right)\Big|_{q_0} \eta_i \eta_j + O(\eta^3)$$

- We are interested in motion around stable equilibrium point q_0,

$$\boxed{V = \frac{1}{2} V_{ij} \eta_i \eta_j} \quad , \quad V_{ij} = \frac{\partial^2 V}{\partial q_i \partial q_j}\Big|_{q_0} > 0$$

$$[V_{ij} = V_{ji}]$$

$\Rightarrow$ For stable motion around

$$\boxed{\mathcal{L} = \frac{1}{2}\left[T_{ij}\dot{\eta}_i\dot{\eta}_j - V_{ij}\eta_i\eta_j\right]}$$

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}_k}\right) - \frac{\partial \mathcal{L}}{\partial q_k} = 0 \equiv \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\eta}_k}\right) - \frac{\partial \mathcal{L}}{\partial \eta_k} = 0$$

$$\boxed{\frac{\partial \mathcal{L}}{\partial \dot{\eta}_k} = \frac{1}{2} T_{ij}\left(\delta_{ik}\dot{\eta}_j + \delta_{jk}\dot{\eta}_i\right)}$$

$$\frac{\partial \mathcal{L}}{\partial \dot{q}_k} = \frac{1}{2}\left(T_{kj}\dot{\eta}_j + T_{ik}\dot{\eta}_i\right)$$

$$= \frac{1}{2}\left[T_{ki}\dot{\eta}_i + T_{ki}\dot{\eta}_i\right] = T_{ki}\dot{\eta}_i$$

$$\frac{\partial \mathcal{L}}{\partial q_k} = -V_{ki}\eta_i$$

$$\boxed{T_{ki}\ddot{\eta}_i + V_{ki}\eta_i = 0}$$

$\eta_i = c a_i e^{-i\omega t}$ → oscillatory function.

≡ We take either real part of this or imaginary part of it, depending on whether we need sine or cosine.

$\boxed{m\ddot{\eta} + k^2\eta = 0}$ → we assume this function can be written as an expansion of $\boxed{\text{sine} / \cos / e^{i\omega t}}$ → Fourier series expansion.

$$\eta = \sum_{k=0}^{\infty} \eta_k e^{-i\omega t}$$

$$-\omega^2 m \eta_0 e^{-i\omega t} + \eta_0 k e^{-i\omega t} = 0$$

$\eta_0 > 0, \; e^{-i\omega t} \neq 0$

$$\Rightarrow -\omega^2 m + k = 0 \Rightarrow \omega^2 = \frac{k}{m} \Rightarrow \omega = \sqrt{\frac{k}{m}}$$

n - coupled eq^ns $\Rightarrow$ linear.

$$-T_{ki} a_i \omega^2 e^{-i\omega t} + \underbrace{V_{ik}}_{=V_{ki}} a_i e^{-i\omega t} = 0$$

$$\Rightarrow -T_{ki} a_i \omega^2 e^{-i\omega t} + V_{ki} a_i e^{-i\omega t} = 0$$

$$\Rightarrow V_{ki} a_i - \omega^2 T_{ki} a_i = 0$$

$$\Rightarrow V \cdot A - \omega^2 \mathbb{I} A = 0$$

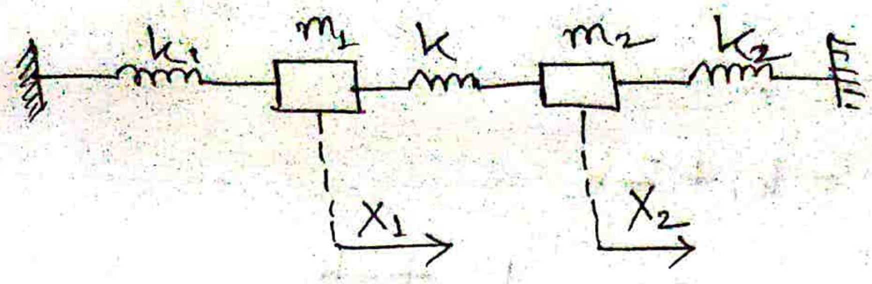
$$\Rightarrow \boxed{(V - \omega^2 \mathbb{I}) A = 0}$$

Similar to eigenvalue problem:

$$(V - \lambda \mathbb{I}) A = 0$$

$\Rightarrow$ non-trivial sol^n exists when

$$\det(V - \lambda \mathbb{I}) = 0$$



▫ $T = \frac{1}{2}(m_1 \dot{x}_1^2 + m_2 \dot{x}_2^2)$

▫ $V = \frac{1}{2} k_1 x_1^2 + \frac{1}{2} k (x_2 - x_1)^2 + \frac{1}{2} k_2 x_2^2$

now, $\eta_1 = x_1$, $\eta_2 = x_2$

$$\boxed{T = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}} , \quad V = \left(\frac{\partial^2 V}{\partial x_i \partial x_j} \right)$$

$$\boxed{V = \begin{pmatrix} k_1 + k & -k \\ -k & k_2 + k \end{pmatrix}}$$

$$\begin{pmatrix} k_1 + k & -k \\ -k & k_2 + k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \omega^2 \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$m_1 = m_2 = m$ $\qquad k_1 = k_2 = k_1$

$$\begin{pmatrix} k_1 + k & -k \\ -k & k_1 + k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} - \omega^2 m \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} k_1 + k - \lambda & -k \\ -k & k_1 + k - \lambda \end{vmatrix} = 0$$

[P.T.O.]

$$\Rightarrow (k_1+k+\lambda)^2 - k^2 = 0$$

$$\lambda = k \; , \quad \lambda = k_1 + 2k$$

$$\omega_1^2 = \frac{k_1}{m} \; , \quad \omega_2^2 = \frac{k_1+2k}{m}$$

$$\begin{pmatrix} k_1+k-k_1 & -k \\ -k & k_1+k-k_1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$= k \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

Putting another eigen value,

$$\begin{pmatrix} -k & -k \\ -k & -k \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0 \rightarrow \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = 0$$

$$\therefore \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

Classical Mechanics :

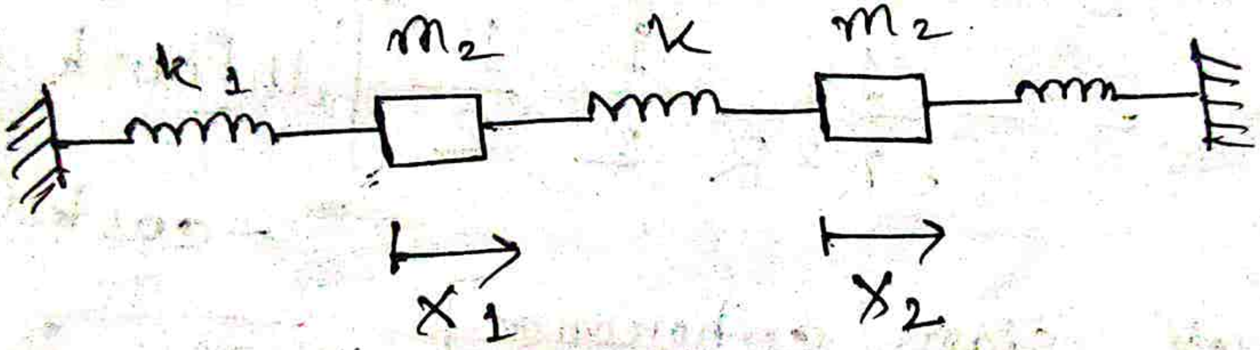
Summary: $\boxed{(V_{ik} - \omega^2 T_{ik})\, a_k = 0}$

- $\omega \rightarrow$ modes of oscillation.

a_k is amplitude of oscillation,

$$\det(V_{ik} - \omega^2 T_{ik}) = 0$$

$$T_{ik} = (\ )\, \delta_{ik}$$



$$m_1 = m_2 = m$$

$$(V_{ik} - \omega^2 m \delta_{ik})\, a_k = 0 .$$

$$m_1 \neq m_2$$

$$T = \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}, \quad V = \begin{pmatrix} k_1 + k & -k \\ -k & k_1 + k \end{pmatrix} .$$

$$\Rightarrow T^{-1} = \begin{pmatrix} \frac{1}{m_1} & 0 \\ 0 & \frac{1}{m_2} \end{pmatrix}$$

$$\therefore \left[\left(V - \omega^2 \cancel{m} T\right)\right] A = 0$$

$$\Rightarrow \left[\begin{pmatrix} \frac{(k_1+k)}{\cancel{m_1}} & -k \\ -k & k_1 + k \end{pmatrix} - \omega^2 \begin{pmatrix} m_1 & 0 \\ 0 & m_2 \end{pmatrix}\right] A = 0$$

$$\Rightarrow \left[\begin{matrix} \frac{(k_1+k)}{m_1} & -\frac{k}{m_2} \\ -\frac{k}{m_1} & \frac{(k_1+k)}{m_2} \end{matrix} \quad - \omega^2 \mathbb{1}\right] A = 0$$

↳ This equation here and,

$$\begin{pmatrix} k_1 + k - m_1\omega^2 & -k \\ k & k_1 + k - m_2\omega^2 \end{pmatrix} = 0$$

give the same result?

- Let us say, hypothetically, we have a matrix, T_{ik} which is not diagonal

$$(V - \omega^2 T) . A = 0$$

- $S^T T S = D \Rightarrow S^T S = I = S S^T$

$$(S^T V - \omega^2 S^T T) S S^T A$$

$$= (S^T V S - \omega^2 S^T S) S^T A$$

$$= (S^T V S - \omega^2 D) S^T A$$

- If V is a positive symmetric matrix than V' is also a positive symmetric matrix, as it is an orthogonal transformation.

$$= [S_1^T V S - \omega^2 D] [A^T S]^T$$

$$= (V' - \omega^2 D) A' = 0$$

$$T = \frac{1}{2} [2\dot{x}_1^2 + 2\dot{x}_1 \dot{x}_2 + 2\dot{x}_2^2]$$

$$V = \frac{1}{2} [3x_1^2 - 2X_1 X_2 + 3X_2^2]$$

[Coupling occurs in both cases]

$\therefore M = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, V = \begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$

$\det |V - \omega^2 M|$

$$= \begin{vmatrix} 3-2\omega^2 & -1-\omega^2 \\ -1-\omega^2 & 3-2\omega^2 \end{vmatrix} = 0$$

$\Rightarrow (3-2\omega^2)^2 - (-1-\omega^2)^2 = 0$

$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - (-(1+\omega^2))^2 = 0$

$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - (1 + 2\omega^2 + \omega^4) = 0$

$\Rightarrow 9 - 12\omega^2 + 4\omega^4 - 1 - 2\omega^2 - \omega^4 = 0$

$\Rightarrow 3\omega^4 - 14\omega^2 + 8 = 0$

$\Rightarrow 3\omega^4 - 12\omega^2 - 2\omega^2 + 8 = 0$

$\Rightarrow 3\omega^2(\omega^2 - 4) - 2(\omega^2 - 4) = 0$

$\Rightarrow (3\omega^2 - 2)(\omega^2 - 4) = 0$

$\therefore \omega = \pm 2, \pm\sqrt{\frac{2}{3}}$

$\therefore \omega^2_{1,2} = 4, \frac{2}{3} \Rightarrow \omega_1^2 = 4, \omega_2^2 = \frac{2}{3}$

Rigid Body Motion (Classical Mechanics)

We are combining N - particle to create 1 ~~p~~ system.

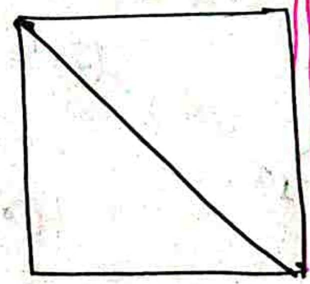
- Rigid body is a system of N - particles in which, distance between any/~~of~~ all two particles remain constant.

$\boxed{r_{ij} = C_{ij}}$ ⇒ r_{ij} is distance from i^{th} particle to the j^{th} particle → constant of motion (imposed one)

Hence, it is a constraint

Distance constraint

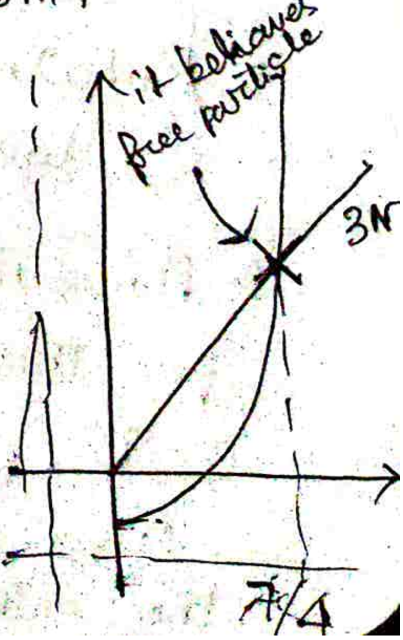
N^2 $r_{ij} = r_{ji}$



$\frac{N(N-1)}{2}$ → no. of independent ~~constraints~~ pairwise ~~coming in~~ distances among n - particles.

- N particles: 3N degrees of freedom.

	$\frac{N(N-1)}{2}$	If free particle	
1	0	3	
2	1	6	
3	3	9	
4	6	12	
5	10	18	



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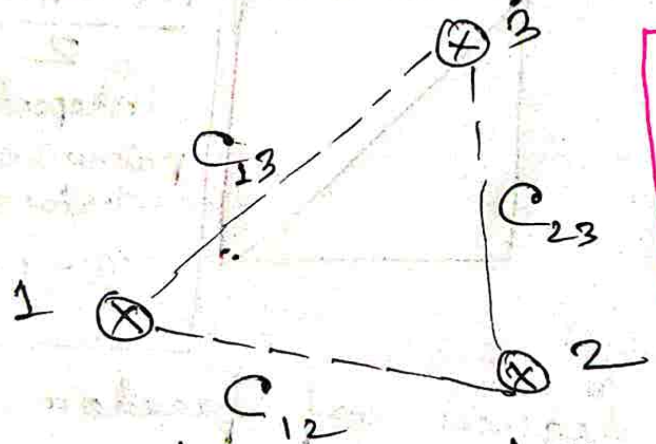
2D

2d • Let us start with particle 1, which is free to move. But particle 2 has to move in a circle of fixed radius (distance) C_{12} from the 2nd particle.

$\boxed{\text{⇒ Position of 1st particle and an angle can completely describe the system.}}$

→ In this frame, we are left with only 1 degree of freedom, that is, rotation about a fixed point.

• Now, we introduce particle 3:



$\boxed{\text{The third particle position remains fixed w.r.t. to positions of particle 1 and 2 given.}}$

• Rotational degree of freedom of particle 2 does not exist anymore.

▷ However, a reflection symmetry exists for this particular 3-particle case in 2-D.

Once we fix 3 points, we essentially have the entire system described.

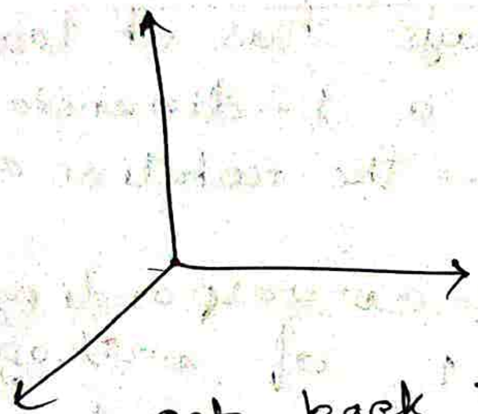
- All non-planar rigid bodies require 6-degrees of freedom in 3 dimensions.

  ↳ 3 degrees out of these are the coordinates of any 1 point of the rigid body.

  → other 3 degrees are angles, rotations about that point.

▷ Which 3 rotations?

Ans.



$$\vec{X} = \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

- If we perform 3 independent rotations, we get back the original ~~spto~~ system.

- Rotation is represented by $X' = AX$.

$$\Rightarrow \boxed{AA^T = A^T A = \mathbb{I}}$$

- Rotation about z-axis:

$\boxed{\vec{\xi} = DX}$, $\boxed{D^T D = \mathbb{I}}$, $\boxed{\hat{z} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}}$

Rotational matrix $D\hat{z} = \hat{z}$ [$\hat{z}$ is an eigenvector of D with eigenvalue 1]

$$\hookrightarrow \begin{pmatrix} \cos\phi & \sin\phi & 0 \\ -\sin\phi & \cos\phi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- 3 eigenvectors for 3-D → 3 axes of rotations are possible.
- eigenvalue equation will be cubic, hence, all eigenvalues might not be real.

- Special cases, eigenvalues ~~is~~ are real.
  Eg: Orthogonal matrices.

- Idea of axes of rotation only makes sense when the dimension is odd.

  Odd dimensions always has at least one fixed direction i.e. a 1-dimensional invariant subspace — the rotation axis.

  ⇒ The eigenvector corresponding to real eigenvalue $+1$ of orthogonal matrix $\mathbb{R}^{2N+1}$ is the axis of rotation.

- Suppose; $R(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$

- has eigenvalues $e^{\pm i\theta}$ → no real eigenvector
  ↓
  no axis!

- Now, $\vec{\xi}' = C\,\vec{\xi}$ about new x-axis by an angle θ.

$$C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{pmatrix}$$

- $X' = B\xi'$ about new z-axis by an angle Ψ.

$$\vec{B} = \begin{pmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

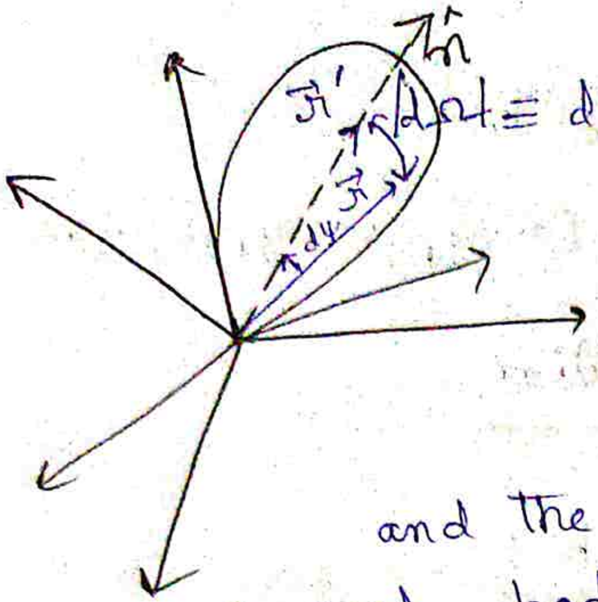
$\Rightarrow \boxed{X' = AX = BCD\,X}$, $\boxed{A = BCD}$

↳ consecutive rotation matrix is also a rotation.

$\boxed{A^T = D^T C^T B^T}$ → Complete set.

$\boxed{DD^T = \mathbb{I}}$, $\boxed{CC^T = \mathbb{I}}$, $\boxed{BB^T = \mathbb{I}}$

$\Rightarrow$ Euler representation of rotation contains full description of rotation in 3-D.



$\boxed{\text{spatial frame}}$

When $t=0$, ~~who~~ there is a fixed frame $\rightarrow$ inertial frame

As time goes, rigid bodies rotate, and the frame rotates with it. (rigid body not rotating w.r.t. this frame). This frame gives the orientation of the rigid body as a function of time.

Let $\hat{\eta}$ be the axes of rotation. $\Psi \equiv$ angle of rotation.

$|d\Omega| \rightarrow d\Psi \quad d\vec{\Omega} = \hat{n}\, d\Psi$

$$\vec{r}\,' = A \cdot \vec{r}\,, \quad A \cdot A^T = \mathbb{I}$$

If $\hat{n} = (0, 0, 1)$

A takes the form: $A = \begin{pmatrix} \cos\Psi & \sin\Psi & 0 \\ -\sin\Psi & \cos\Psi & 0 \\ 0 & 0 & 1 \end{pmatrix}$

• For small rotation; $\Psi \rightarrow d\Psi$ $\quad A = \mathbb{I} + \begin{pmatrix} 0 & d\Psi & 0 \\ -d\Psi & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \epsilon$

$$\Rightarrow A = \begin{pmatrix} 1 & d\Psi & 0 \\ -d\Psi & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$\epsilon \rightarrow$ antisymmetric matrix.

$$\mathbb{A} = (\mathbb{I} + \epsilon), \text{ Assume elements of } \epsilon_{ij} << 1$$

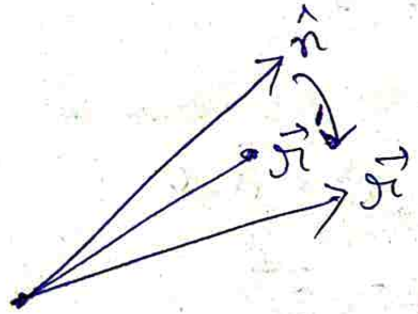
$$\epsilon_{ij}\,\epsilon_{ji} \simeq 0$$

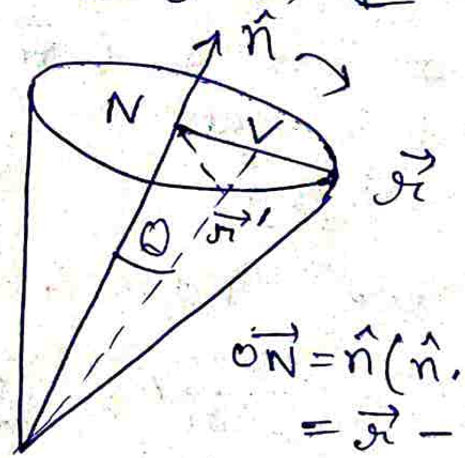
Now, $$A.A^T = \mathbb{I} = (\mathbb{I} + \epsilon)(\mathbb{I} + \epsilon)^T$$

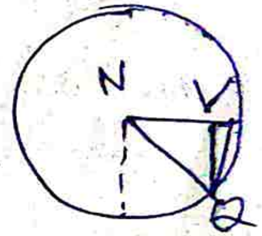
$$= \mathbb{I} + \epsilon + \epsilon^T + \cancel{\epsilon\epsilon^T} \rightarrow 0$$

$$\therefore \mathbb{I} + \epsilon + \epsilon^T = \mathbb{1}$$

$$\Rightarrow \epsilon + \epsilon^T = 0 \Rightarrow \epsilon = -\epsilon^T$$



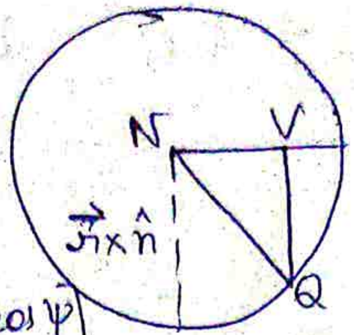




$$\vec{ON} = \hat{n}(\hat{n}.\vec{r}) \quad \vec{NP}$$
$$= \vec{r} - \hat{n}(\hat{n}.\vec{r})$$
$$a' =$$

Let us take a vector $\hat{n} \times \vec{r}$

$$\vec{r}\,' = \overrightarrow{ON} + \overrightarrow{NQ}$$
$$= \overrightarrow{ON} + \overrightarrow{NQ} + \overrightarrow{VQ}$$



$$\overrightarrow{NV} = [r - \hat{n}\,(\vec{r}\cdot\hat{n})\cos\psi]$$

$$VQ = (\vec{r} \times \hat{n}) \sin\psi$$

$$\vec{r} = \vec{r}\cos\Psi + \hat{n}\,(\hat{n}\cdot\vec{r})\,(1-\cos\Psi) + (\vec{r}\times\hat{n})\sin\Psi$$

$$\vec{r}\,' = r + (\vec{r}\times\hat{n})\,d\Psi$$

We define a vector:

$$d\vec{\Omega} = d\Psi\,\hat{n}$$

Then we have: $\vec{r}\,' = \vec{r} \times d\Omega$

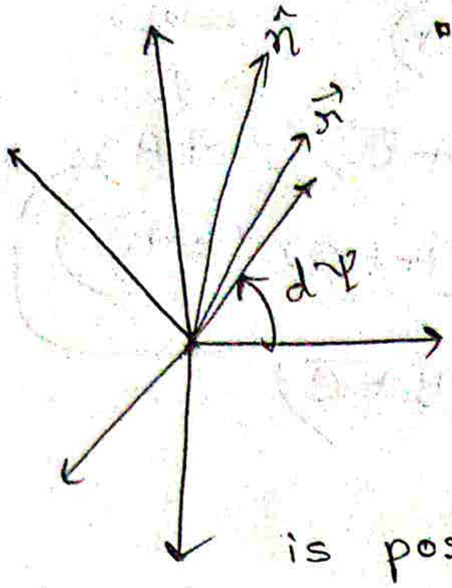
$$\vec{r}\,' = (\mathbb{I} + \epsilon)\,\vec{r}$$

$$\vec{r} = (x_1, x_2, x_3);\ d\Omega = (d\Omega_1, d\Omega_2, d\Omega_3)$$

$$\vec{r} \times d\Omega = \hat{i}(x_2\, d\Omega_3 - x_3\, d\Omega_2)$$
$$+ \hat{j}(-x_1\, d\Omega_3 + x_2\, d\Omega_1)$$
$$+ \hat{k}(x_1 d\Omega_2 - x_2\, d\Omega_1)$$

$$\begin{pmatrix} 0 & d\Omega_3 & -d\Omega_3 \\ -d\Omega_2 & 0 & d\Omega_1 \\ d\Omega_2 & -d\Omega_1 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} + \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} =$$

Classical Mechanics : 19/11/2025



- A frame that is rotating with an arbitrary axis of rotation ($\hat{n}$) and whose origin coincides with the origin of the spatial frame. $\vec{r}$ is position vector w.r.t. the spatial frame. $\vec{r}\,'$ is position vector w.r.t. rotating frame.

- $\vec{r}\,' = \vec{r} + \vec{r} \times \hat{n}\, d\Psi$, $d\vec{\Omega} = \hat{n}\, d\Psi$
- $\vec{r}\,' \simeq \vec{r} + \epsilon\, \vec{r}$

$$\epsilon = \begin{pmatrix} 0 & -d\Omega_3 & d\Omega_2 \\ d\Omega_2 & 0 & -d\Omega_1 \\ -d\Omega_2 & d\Omega_1 & 0 \end{pmatrix}$$

- This rotation is to bring in the anti-symmetric rotation matrix.

- $(\vec{r} \times \hat{n})_i = \epsilon_{ijk}\, r_j\, n_k$
- $(\hat{n} \times \vec{r})_i = \epsilon_{ijk}\, n_j\, r_k\, d\Psi$

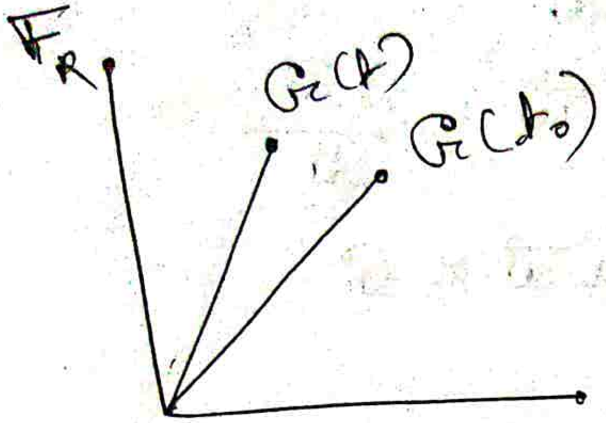
$$= \epsilon_{ijk}\, d\Omega_j\, r_k$$

$d\Omega_i = d\Psi n_i$

$$= \epsilon_{kij}\, d\Omega_j\, r_k$$

$d\Omega_i = d\Psi n_i$, $\epsilon_{kij}\, d\Omega_j = \epsilon_{ki}$

Let G be an arbitrary vector. We see it w.r.t. to the two frames.



($F_R \equiv$ Rotating Frame attached to earth)

If I am ~~on~~ moving on earth, my position is given by $\vec{G}$. But if I am moving around, $\vec{G}$ changes as a function of time.

$$G_S = G_R + dG_{rot}$$

$$d\vec{G} = d\vec{\Omega} \times \vec{G}$$

When the vector is transformed, the vector looks different due to the transformation.

$$dG_S = G_S(t) - G(t_0) = G_R(t) - G_R(t_0)$$

$$dG_S = dG_R + dG_{rot}$$

$$dG_S = dG_R + \underline{d\Omega \times G}$$

↳ How earth is rotating w.r.t. to that frame [When I am fixed w.r.t. to the inertial frame]

$$\lim_{t\to 0}\frac{dr_s}{dt} = \frac{dr_R}{dt} + \frac{d\Omega}{dt}\times r$$

$$\Rightarrow V_s = V_R + \vec{\omega}\times\vec{r} \Rightarrow \vec{\omega} = \frac{d\vec{\Omega}}{dt}$$

- $$\left.\frac{dG}{dt}\right|_s = \left.\frac{dG}{dt}\right|_r + \vec{\omega}\times G.$$

- $$\frac{dV_s}{dt} = \left.\frac{dV_s}{dt}\right|_r + \vec{\omega}\times\vec{V}_s$$

$$\frac{dV_s}{dt} = \frac{d}{dt}\left(\vec{V}_R + \vec{\omega}\times\vec{r}\right) + \vec{\omega}\times\left(\vec{V}_R + \vec{\omega}\times\vec{r}\right)$$

$$= \frac{d\vec{V}_R}{dt} + \vec{\omega}\times\vec{r} + 2\vec{\omega}\times\vec{V}_R + \vec{\omega}\times\vec{\omega}\times\vec{r}$$

$$\Rightarrow a_0 = a_R + \vec{\omega}\times\vec{r} + 2\vec{\omega}\times\vec{V}_R + \vec{\omega}\times\vec{\omega}\times\vec{r}$$

The moment we take w ~~bd~~ to be constant, the whole thing becomes pointless (Why?)

→ Here, $\vec{\omega}\times\vec{r} \longrightarrow$ Euler Force

$2\vec{\omega}\times\vec{V}_R \longrightarrow$ Coriolis Force

$\vec{\omega}\times\vec{\omega}\times\vec{r} \rightarrow$ Centrifugal Force.

- $dG = d\Omega \times G \equiv \epsilon G$

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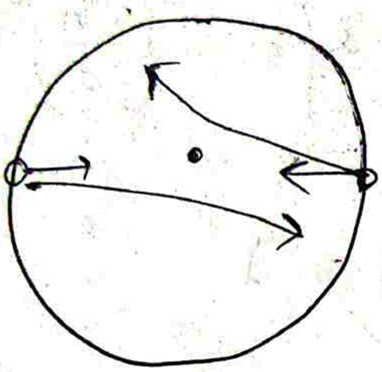
- If $\omega =$ constant, coriolis force disappears.
- In north and south hemispheres, velocities in opposite directions.



• → axis of rotation.

If ~~we~~ the two people were sitting diametrically opposite positions on a merry-go-round and spray water at each other, it will get deflected and will not hit the other person.